

Linear Algebra II

Coordinates, Determinant, Eigenstructure

Mathematics for Robotics · ROB-GY 6013 · NYU Tandon

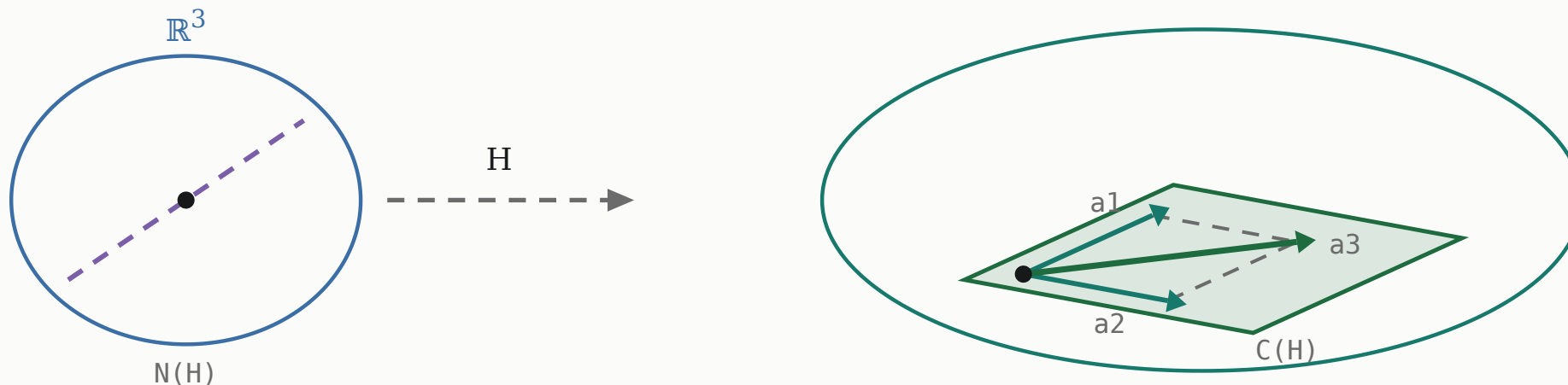
Jing Zhang · Fall 2026 · September 9

One Map, Many Rulers

$$H = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

RANK-NULLITY $\dim \ker H + \dim \operatorname{im} H = 1 + 2 = 3$

$\operatorname{im} H$ the outputs' plane, $\mathcal{C}(H)$



$$\mathcal{N}(A) = \ker(A) = \{x \in \mathbb{R}^n : Ax = 0\}$$

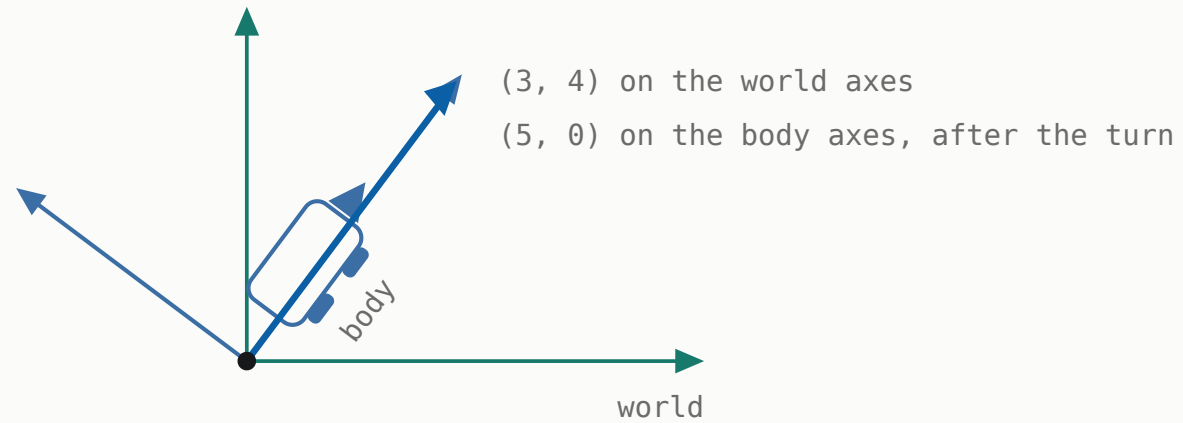
$$\mathcal{N}(H) = \operatorname{span}\{[-1, -1, 1]^T\}$$

$\mathcal{N}(A) = \ker(A)$ the inputs A sends to 0

$x \in \mathbb{R}^n$ an input column of n reals

span all its multiples: a line

One Map, Many Rulers



Which choice makes the matrix simple?

One Map, Many Rulers

1 · COORDINATES

what are these numbers
written in?

2 · ROTATIONS

which change of basis
is a turn?

3 · DETERMINANT

does it collapse,
and does it flip?

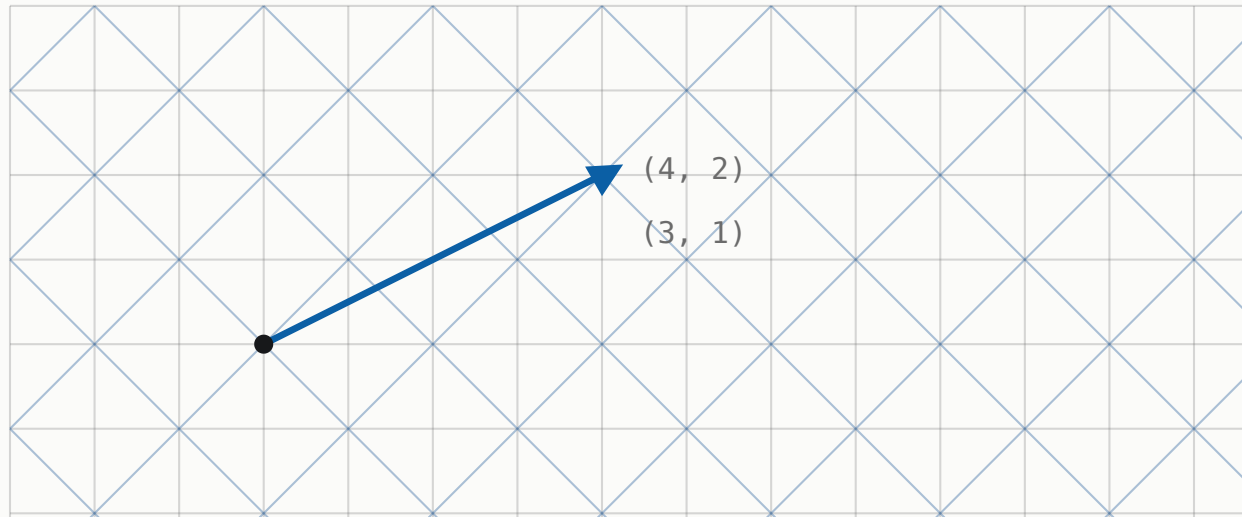
4 · EIGENVECTORS

which directions
survive?

5 · DIAGONALIZATION

what is A to the k ?

Coordinates and Change of Basis



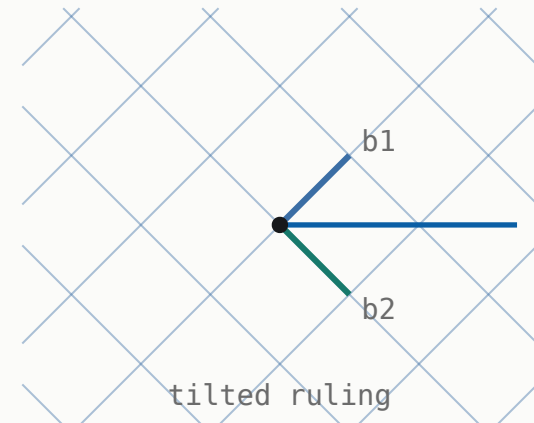
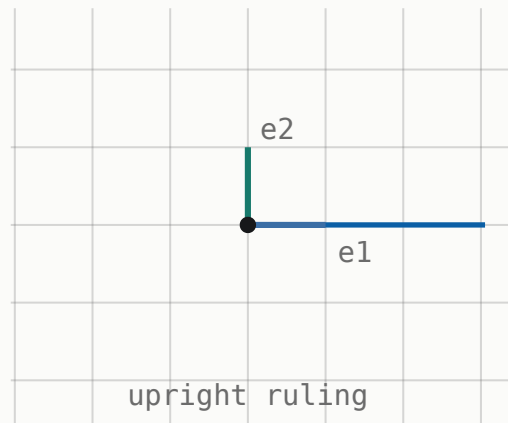
Coordinates and Change of Basis

$\mathcal{B} = (b_1, \dots, b_n)$ is a basis of $\mathbb{R}^n \iff$ independent and spans $\mathbb{R}^n$

independent none of them is a mixture of the rest

spans every vector is some mixture of the list

n a basis of $\mathbb{R}^n$ has exactly n vectors



Axler, S. *Linear Algebra Done Right*, 4th ed., 2.26 basis, p. 39. Springer, 2024. Open access.

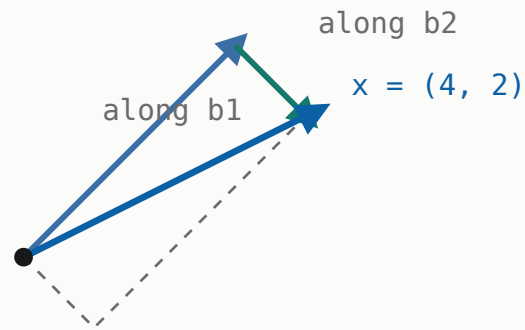
Coordinates and Change of Basis

ONE RECIPE ONLY $x = \alpha_1 b_1 + \cdots + \alpha_n b_n$ in exactly one way

α_j the amount of b_j the recipe calls for

exists because the list spans

only one because the list is independent



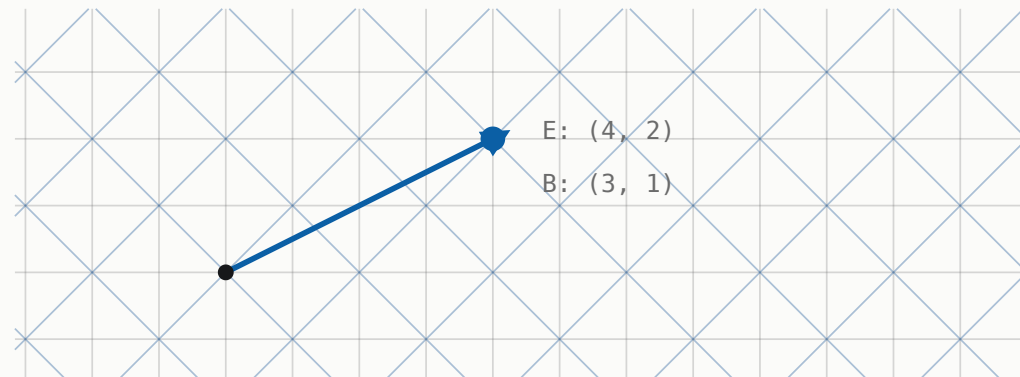
Axler, S. *Linear Algebra Done Right*, 4th ed., 2.28 criterion for basis, p. 39. Springer, 2024. Open access.

Coordinates and Change of Basis

COORDINATE VECTOR

$$[x]_{\mathcal{B}} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \text{ where } x = \alpha_1 b_1 + \cdots + \alpha_n b_n$$

- × the arrow itself, out in the world
- $[x]_{\mathcal{B}}$ a vector in $\mathbb{R}^n$ holding the coefficients, not the arrow x
- α_j how much of b_j the one recipe uses



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.73 the matrix of a vector, p. 88. Springer, 2024. Open access.

Coordinates and Change of Basis

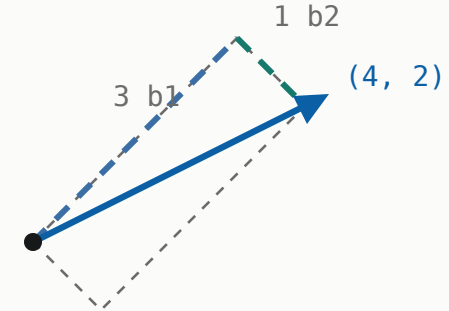
WORKED EXAMPLE

$[x]_{\mathcal{E}} = (4, 2)^T$, $b_1 = (1, 1)$, $b_2 = (1, -1)$. Then $\alpha_1 + \alpha_2 = 4$ and $\alpha_1 - \alpha_2 = 2$, so $\alpha_1 = 3$, $\alpha_2 = 1$. Check: $3b_1 + b_2 = (4, 2)$.

$$[x]_{\mathcal{E}} = (4, 2)^T \longrightarrow [x]_{\mathcal{B}} = (3, 1)^T$$

$[x]_{\mathcal{E}}$ the given standard-coordinate column

$[x]_{\mathcal{B}}$ the same vector in $\mathcal{B}$: three b_1 , one b_2

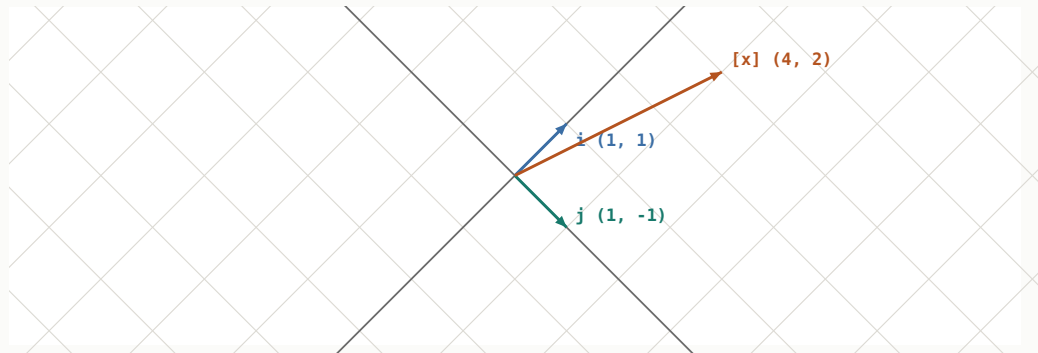


Coordinates and Change of Basis

CONVERT TO STANDARD COORDINATES $\boxed{[x]_{\mathcal{E}} = B [x]_{\mathcal{B}}}, \quad B = \begin{bmatrix} [b_1]_{\mathcal{E}} & \cdots & [b_n]_{\mathcal{E}} \end{bmatrix}$

B columns are the basis vectors in standard coordinates

$[x]_{\mathcal{E}}$ written x once the standard basis is understood



$$3 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{b1} + 1 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}_{b2} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}_{[x] \text{ in } E}$$

Coordinates and Change of Basis

CONVERT BACK WITH THE INVERSE

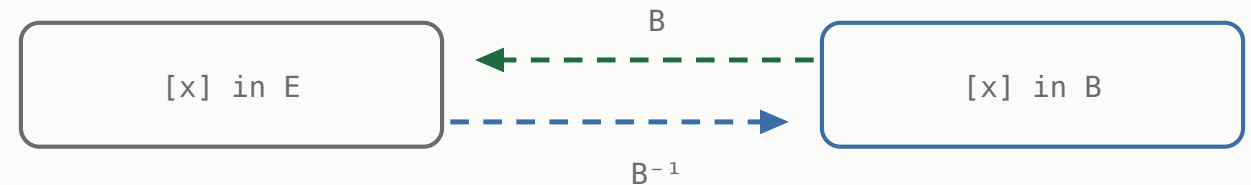
$$[x]_{\mathcal{B}} = B^{-1}[x]_{\mathcal{E}}$$

B^{-1} undoes B , back to $\mathcal{B}$ -coordinates

WORKED EXAMPLE

$M^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$: swap, negate,
divide by $\det M$.

$B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $\det B = -2$, so $B^{-1} = \frac{1}{2}B$
and $[x]_{\mathcal{B}} = \frac{1}{2}(6, 2)^T = (3, 1)^T$.



Coordinates and Change of Basis

$$A^{-1}A = AA^{-1} = I$$

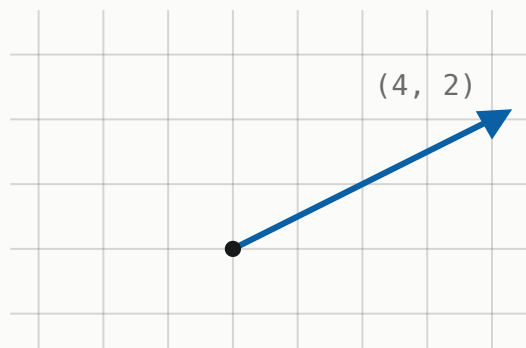
STANDARD BASIS

$$e_1, \dots, e_n : \quad B = I \Rightarrow [x]_{\mathcal{E}} = I^{-1}x = x$$

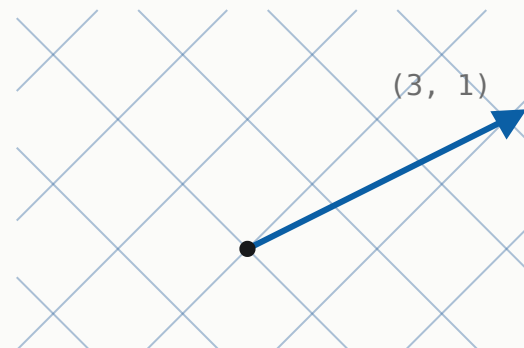
A^{-1} undoes A from either side

e_j 1 in slot j , 0 everywhere else

I its basis matrix, and its own inverse



$B = I$



$B = [b_1 \ b_2]$

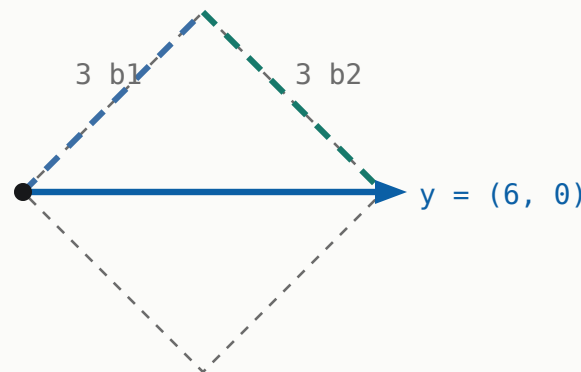
PRACTICE

Coordinates and Change of Basis

Same $\mathcal{B} = \{(1, 1), (1, -1)\}$. What is $[y]_{\mathcal{B}}$ for $y = (6, 0)$?

$$[y]_{\mathcal{B}} = (3, 3)^T$$

Check it forwards: $3(1, 1) + 3(1, -1) = (6, 0)$. Or read it off:
 $B^{-1}(6, 0) = \frac{1}{2}(6, 6) = (3, 3)$.

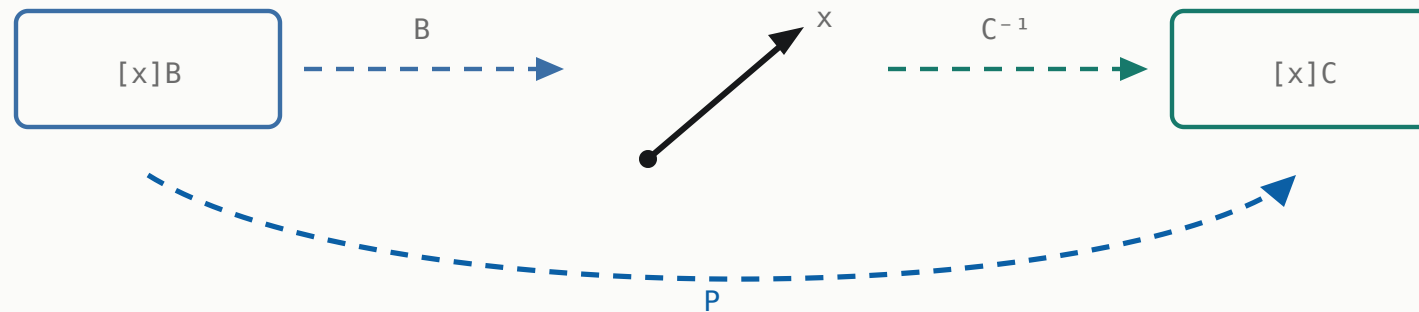


Coordinates and Change of Basis

$$[x]_C = P_{C \leftarrow B} [x]_B, \quad P_{C \leftarrow B} = C^{-1}B$$

P turns B-coordinates into C-coordinates

$C^{-1}B$ rebuild the vector from B, then read it off in C



Axler, S. **Linear Algebra Done Right**, 4th ed., 3.84 change-of-basis formula, p. 93. Springer, 2024. Axler writes this matrix as $M(I, \dots)$, the identity map read between two bases.

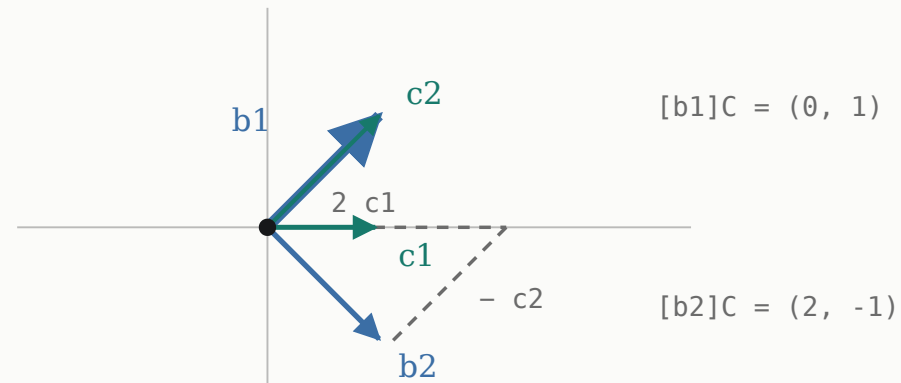
$$B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Coordinates and Change of Basis

WORKED EXAMPLE

$$\mathcal{C} = \{(1, 0), (1, 1)\}, \text{ so } C^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \text{ and } P = C^{-1}B = \begin{pmatrix} 0 & 2 \\ 1 & -1 \end{pmatrix}.$$

$b_1 = 0c_1 + 1c_2$ and $b_2 = 2c_1 - 1c_2$: the columns of P .



Coordinates and Change of Basis

$$x = (4, 2) \text{ in } \mathcal{E}, \quad [x]_{\mathcal{B}} = (3, 1), \quad [x]_{\mathcal{C}} = (2, 2)$$

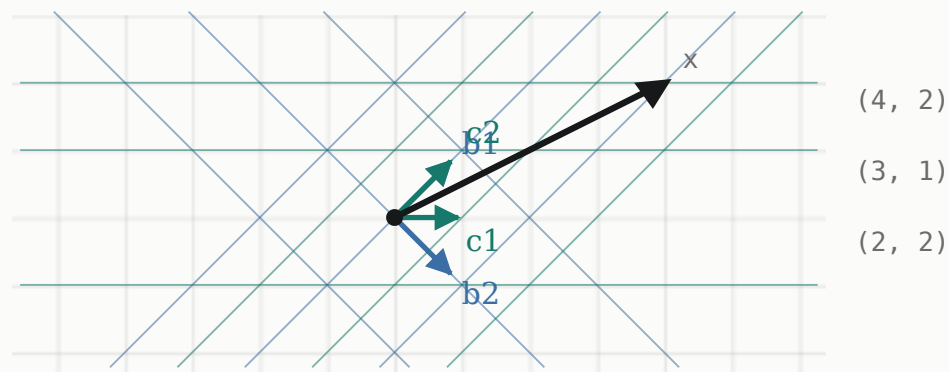
(4,2) read in the standard basis $\mathcal{E}$

(3,1) read in $\mathcal{B} = \{(1,1), (1,-1)\}$

(2,2) read in $\mathcal{C} = \{(1,0), (1,1)\}$

WORKED EXAMPLE

$$P[x]_{\mathcal{B}} = \begin{pmatrix} 0 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$



$$P = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix}$$

Coordinates and Change of Basis

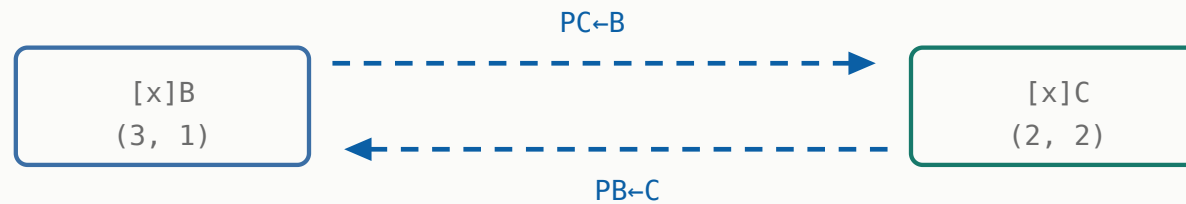
$$P_{B \leftarrow C} = P_{C \leftarrow B}^{-1} = B^{-1}C$$

left side C-coordinates in, B-coordinates out
 $B^{-1}C$ rebuild from C, then read off in B

$$P^{-1} = \begin{pmatrix} 1/2 & 1 \\ 1/2 & 0 \end{pmatrix}, \quad P^{-1}(2, 2) = (3, 1)$$

(2,2) the C-column

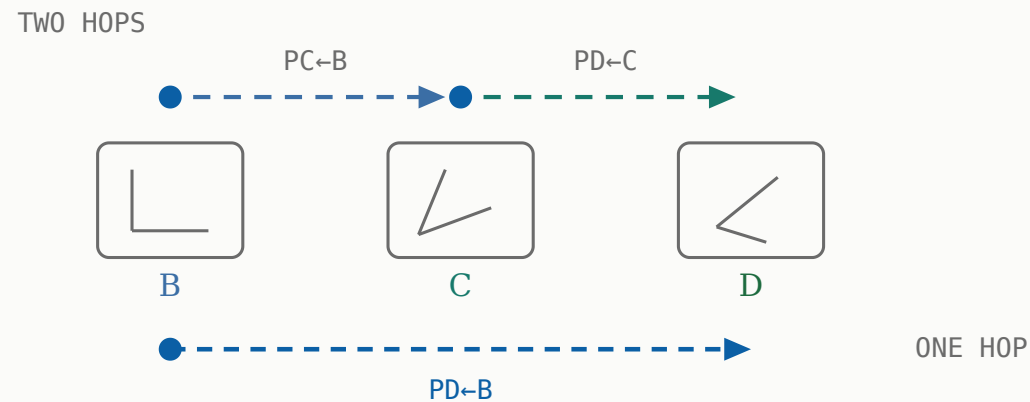
(3,1) back to the B-column



Coordinates and Change of Basis

$$P_{\mathcal{D} \leftarrow \mathcal{B}} = P_{\mathcal{D} \leftarrow \mathcal{C}} P_{\mathcal{C} \leftarrow \mathcal{B}}$$

inner pair C meets C and cancels
order the right factor acts first



Strang, G. **18.06 Linear Algebra**, Lecture 31, change of basis. MIT OpenCourseWare, 2010. CC BY-NC-SA 4.0.

Coordinates and Change of Basis

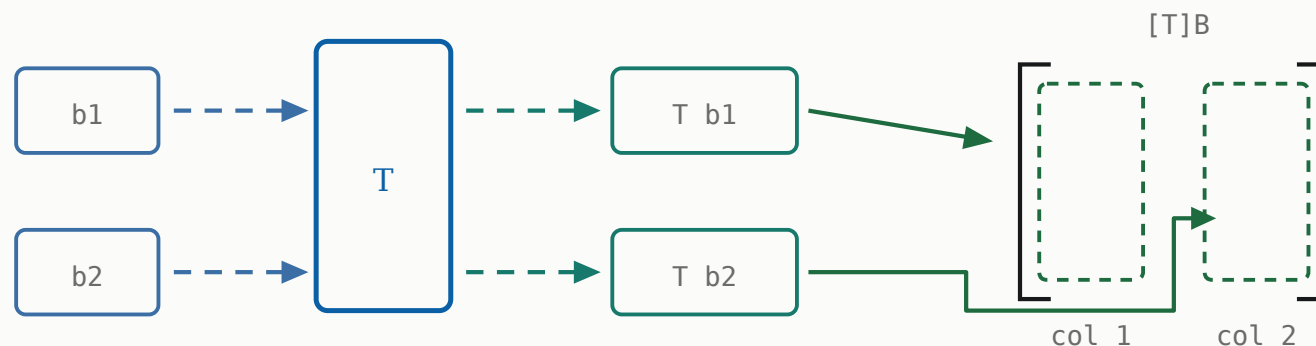
$$[Tx]_{\mathcal{B}} = [T]_{\mathcal{B}} [x]_{\mathcal{B}}, \quad \text{column } j = [Tb_j]_{\mathcal{B}}$$

T a linear map from the space to itself

$[x]_{\mathcal{B}}$ coordinates in, coordinates out, same basis

column j feed in b_j , write the answer in the same basis

$[T]_{\mathcal{B}}$ depends on T and on the ordered basis $\mathcal{B}$



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.31 matrix of a linear map $M(T)$, p. 69. Springer, 2024.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Coordinates and Change of Basis

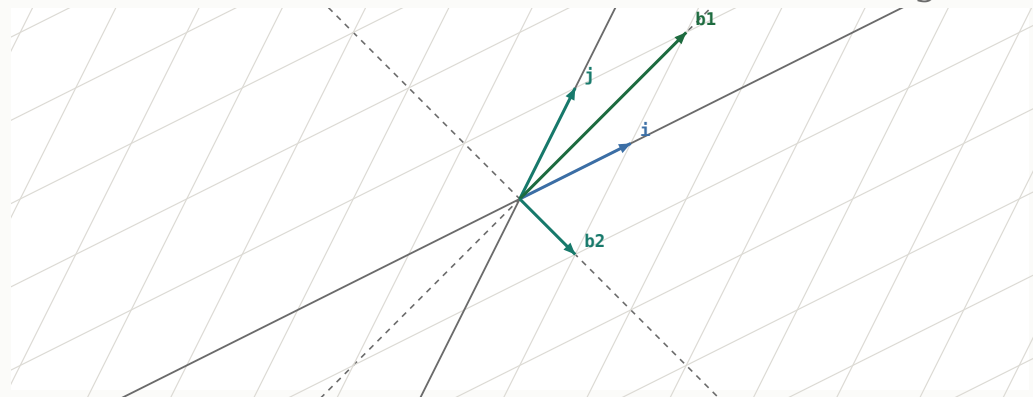
WORKED EXAMPLE

$Ab_1 = (3, 3) = 3b_1 + 0b_2$, so column 1 is $(3, 0)^T$. $Ab_2 = (1, -1) = 0b_1 + 1b_2$, so column 2 is $(0, 1)^T$.

$$[A]_{\mathcal{B}} = B^{-1}AB = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$$

$B^{-1}AB$ rebuild, apply A , read off

3 and 1 the factor each basis direction is stretched by
off-diagonal zero here: the two directions no longer mix



Coordinates and Change of Basis

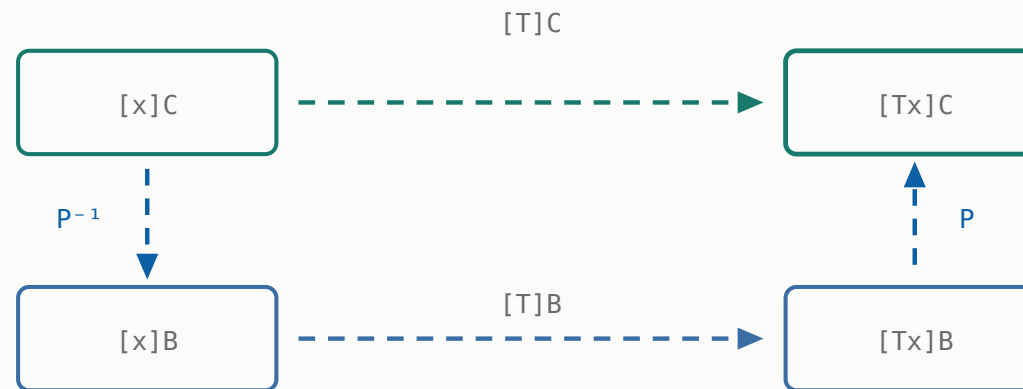
$$[T]_{\mathcal{C}} = P [T]_{\mathcal{B}} P^{-1}, \quad P = P_{\mathcal{C} \leftarrow \mathcal{B}}$$

P puts B-coordinates into C, the same P as before

$[T]_{\mathcal{B}}$ the operator acts, in $\mathcal{B}$

P^{-1} takes the C-column into B, and acts first

similar the name for two matrices related this way



Axler, S. *Linear Algebra Done Right*, 4th ed., 3.84 change of basis, p. 93. Springer, 2024. The word **similar** is Strang's: MIT 18.06, Lecture 28.

$$C = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Coordinates and Change of Basis

WORKED EXAMPLE

$$C^{-1}A = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}, \text{ then } (C^{-1}A)C = \begin{pmatrix} 1 & 0 \\ 1 & 3 \end{pmatrix} = [A]_C.$$

$$\text{tr}(A) = \sum_i a_{ii}$$

$$\text{tr}(B^{-1}AB) = \text{tr}(A)$$

tr sum of the diagonal entries

$B^{-1}AB$ new basis, same sum

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$2 + 2 = 4$$

$$[A]_C = \begin{bmatrix} 1 & 0 \\ 1 & 3 \end{bmatrix}$$

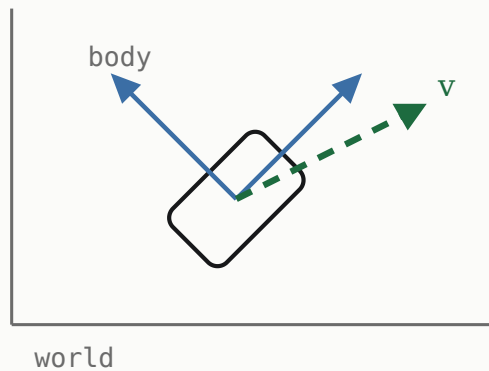
$$1 + 3 = 4$$

$$[A]_B = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$3 + 1 = 4$$

Coordinates and Change of Basis

The same gains, written in two frames



IN THE WORLD FRAME

$$\begin{bmatrix} 0.7 & 0.2 \\ 0.2 & 0.7 \end{bmatrix}$$

IN THE BODY FRAME

$$\begin{bmatrix} 0.9 & 0 \\ 0 & 0.5 \end{bmatrix}$$

Axler, S. *Linear Algebra Done Right*, 4th ed., 3.84 change-of-basis formula, p. 93. Springer, 2024. Open access.

Coordinates and Change of Basis

$[T]_{\mathcal{B}} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ with $\mathcal{B} = \{(1, 1), (1, -1)\}$. What is $[T]_{\mathcal{E}}$?

$$\begin{array}{ccc} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} & \xrightarrow{B (\cdot) B^{-1}} & \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \\ \text{IN } \mathcal{B} & & \text{IN } \mathcal{E} \end{array}$$

$$[x]_{\mathcal{B}} = B^{-1}[x]_{\mathcal{E}}$$

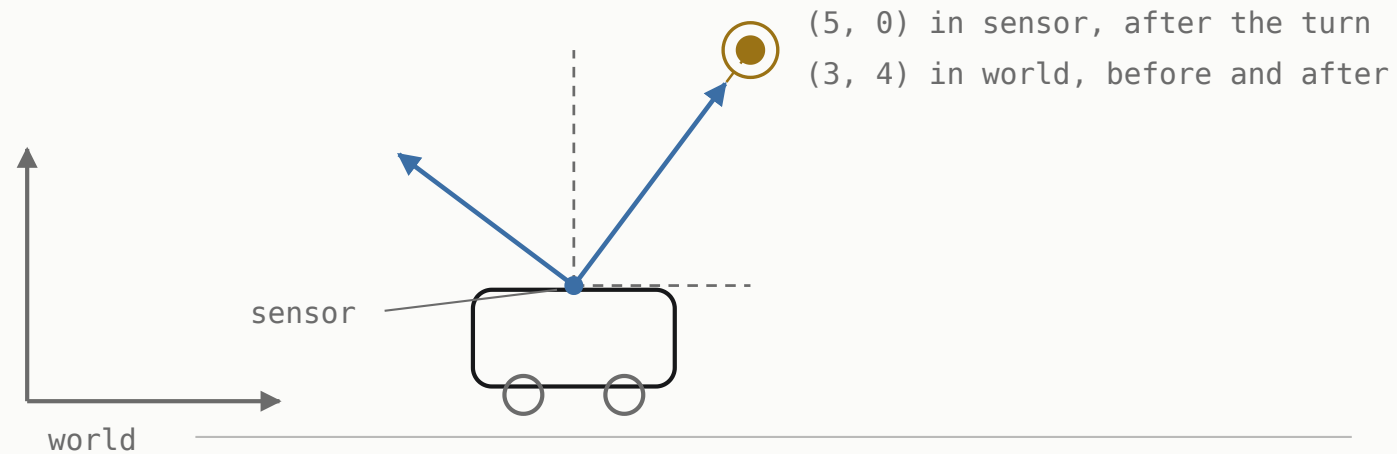
$$[Tx]_{\mathcal{B}} = [T]_{\mathcal{B}}[x]_{\mathcal{B}}$$

$$[Tx]_{\mathcal{E}} = B[Tx]_{\mathcal{B}}$$

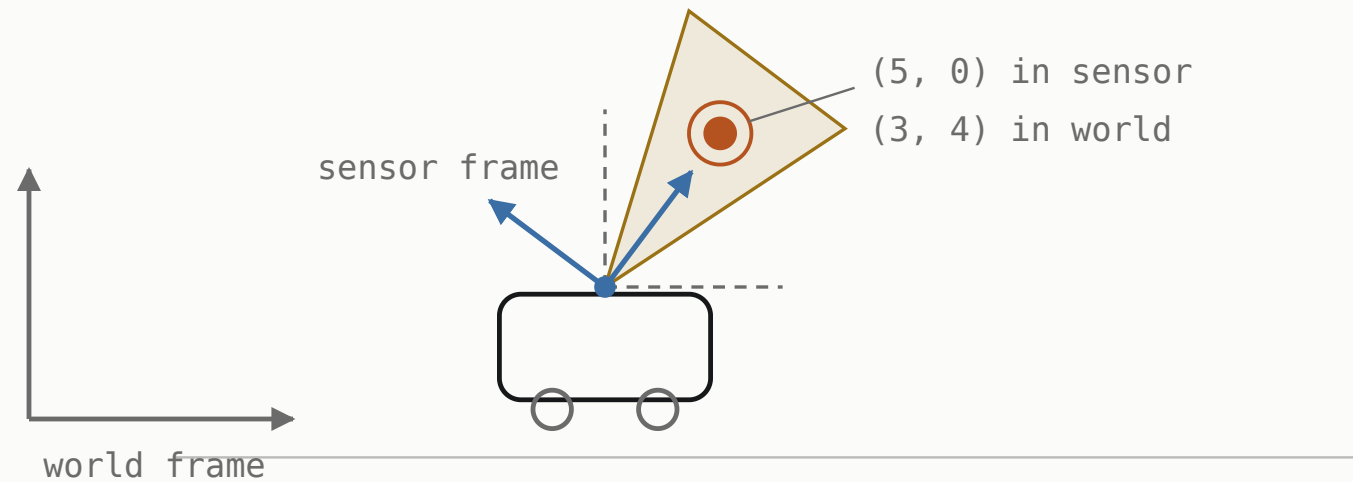
$$\boxed{[T]_{\mathcal{E}} = B[T]_{\mathcal{B}}B^{-1}}$$

the three lines into B , act, back out; together they are the box

Rotations and Frames



Rotations and Frames



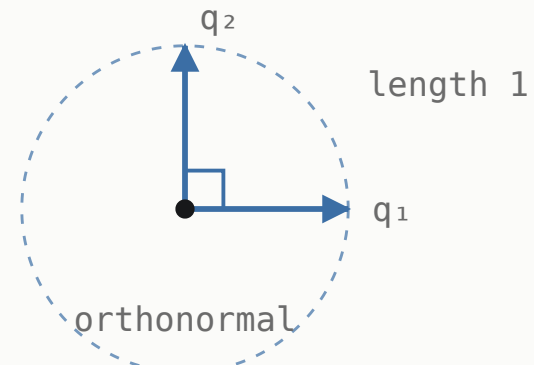
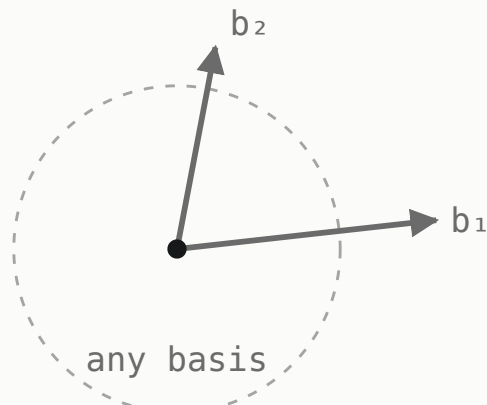
Rotations and Frames

$$\text{ORTHONORMAL} \quad \langle q_i, q_j \rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

q_i the basis vectors, one for each axis

$i = j$ every one of them has length one

$i \neq j$ every pair meets at a right angle



Axler, S. *Linear Algebra Done Right*, 4th ed., 6.22 orthonormal, p. 197; 6.27 orthonormal basis, p. 199. Springer, 2024. Open access.

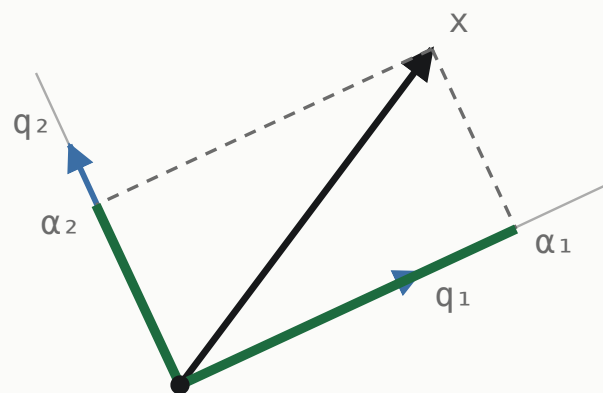
Rotations and Frames

$$[x]_{\mathcal{Q}} = Q^T x, \quad \text{i.e. } \alpha_j = \langle x, q_j \rangle$$

Q the $n \times n$ matrix with columns $q_1, \dots, q_n$

Q^T exactly Q^{-1} : a transpose, nothing to solve

α_j how far x reaches along axis j



Axler, S. **Linear Algebra Done Right**, 4th ed., 6.30(a) writing a vector as a linear combination of an orthonormal basis, p. 200. Springer, 2024. Open access.

Rotations and Frames

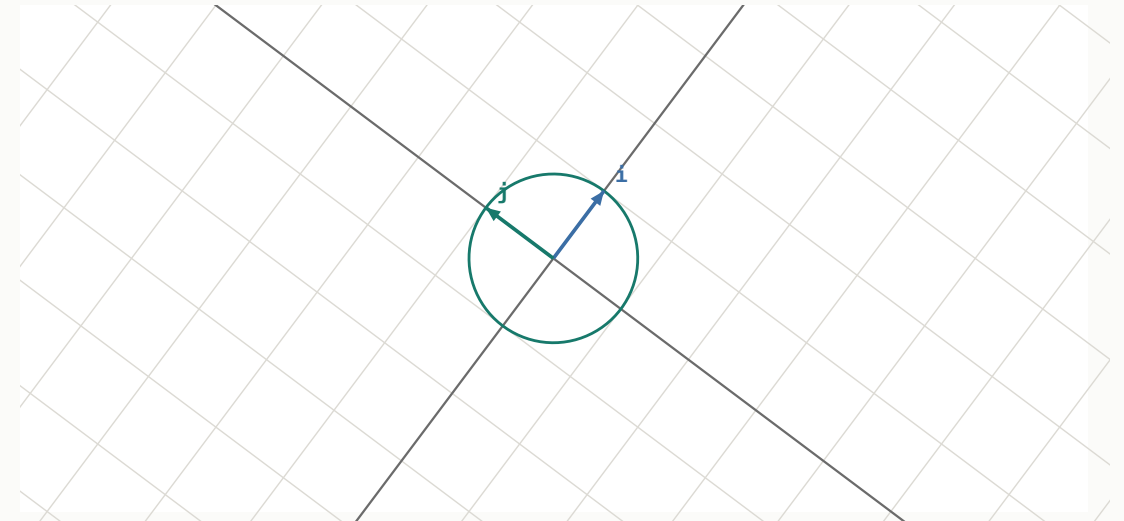
ORTHOGONAL $Q^T Q = I \iff$ the columns of Q are orthonormal, $Q^{-1} = Q^T$

$(Q^T Q)_{ij}$ column i dotted with column j

I ones on the diagonal, zeros off it

$Q^{-1} = Q^T$ the inverse costs one transpose

$$Q^T Q = \begin{matrix} & \begin{matrix} q_1 & q_2 & q_3 \end{matrix} \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \end{matrix} & \begin{bmatrix} \boxed{1} & 0 & 0 \\ 0 & \boxed{1} & 0 \\ 0 & 0 & \boxed{1} \end{bmatrix} \end{matrix} = I$$



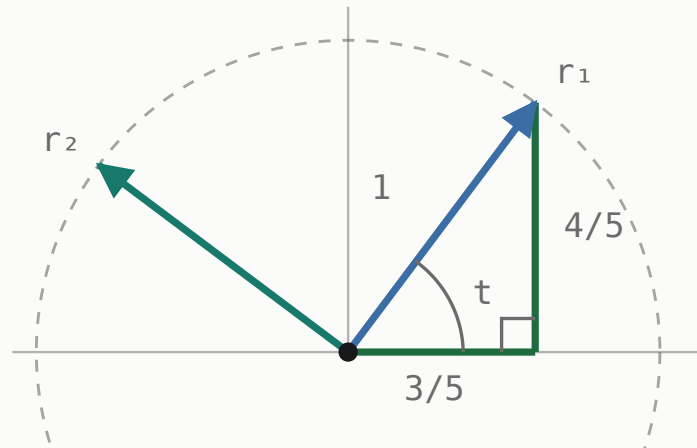
Both ends are orthogonal; the frames between them are smaller.

Grizzle, J. W. **ROB 501: Mathematics for Robotics**, notes 11, symmetric and orthogonal matrices. University of Michigan, 2018. CC BY-NC 4.0.

Rotations and Frames

WORKED EXAMPLE

$R = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$. Column one with itself: $\frac{9}{25} + \frac{16}{25} = 1$. Column one with column two: $\frac{-12}{25} + \frac{12}{25} = 0$. So $R^T R = I$, with $\cos t = \frac{3}{5}$, $\sin t = \frac{4}{5}$, $t \approx 53.13^\circ$.



Rotations and Frames

ROTATION GROUP $SO(n) = \{R \in \mathbb{R}^{n \times n} : R^T R = I, \det R = 1\}$

$R^T R = I$ lengths and angles survive the map

$\det R = 1$ a turn, not a mirror; the number itself arrives next chapter

n 2 for planar orientations, 3 for spatial ones

	$R^T R = I$	$\det R$
rotation	✓	☐
reflection	✓	☐ → ch. 3
scaling $\times 2$	✗	☐

Lynch, K. M. & Park, F. C. **Modern Robotics**, Definitions 3.1 and 3.2, p. 70. Cambridge, 2017. $SO(3)$ and $SO(2)$ are defined separately there; the general n is our shorthand.

$$R = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}$$

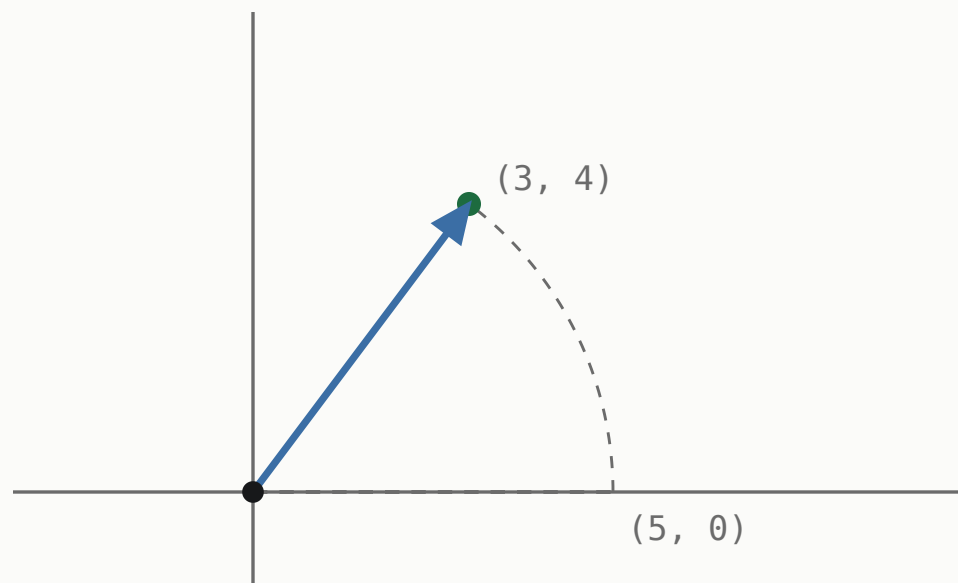
Rotations and Frames

ACTIVE $x' = Rx :$ $R(5, 0) = (3, 4)$

x the arrow before the turn

R the turn itself, the same for every arrow

x' a different arrow, of the same length



Rotations and Frames

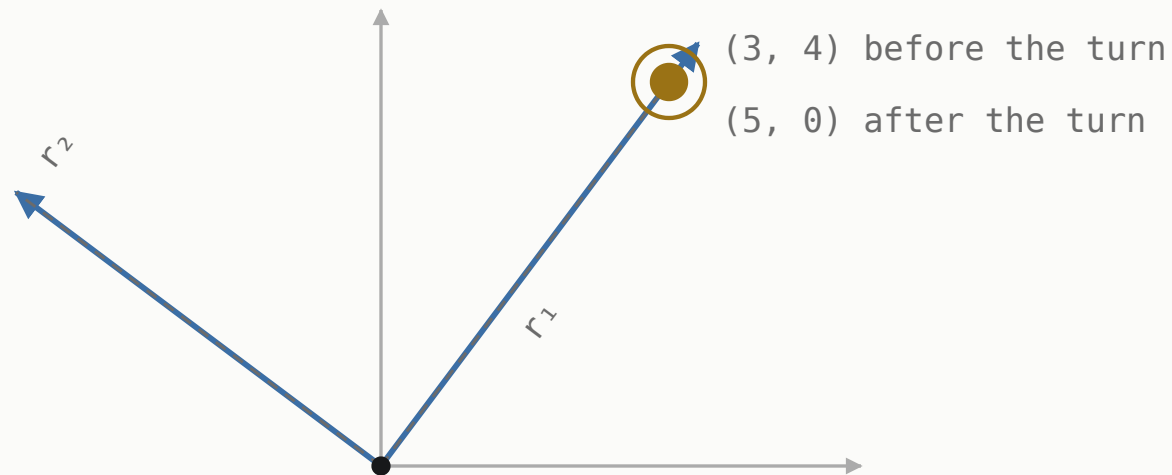
$$R = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}$$

PASSIVE $[x]_{\mathcal{R}} = R^T x : \quad R^T(3, 4) = (5, 0)$

x one physical point; nothing about it moves

R^T chapter 1's $B^{-1}x$, with $B = R$

$[x]_{\mathcal{R}}$ the same point's new address



Lynch, K. M. & Park, F. C. **Modern Robotics**, §3.2.1.2, uses of rotation matrices, p. 71. Cambridge University Press, 2017.

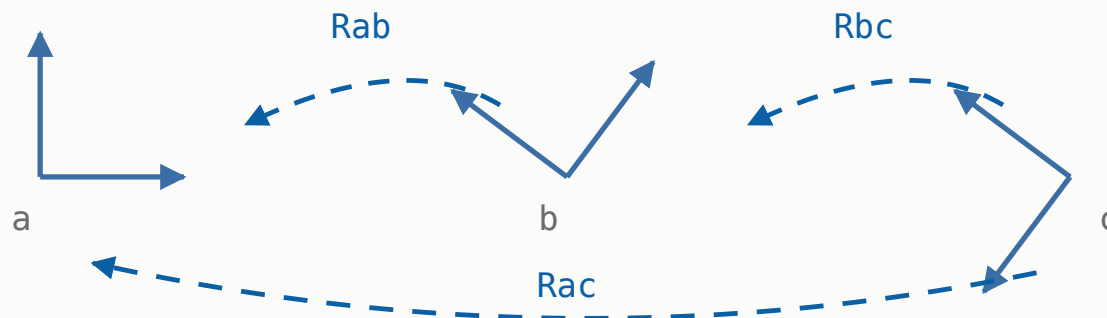
Rotations and Frames

COMPOSE $R_{ac} = R_{ab}R_{bc}$

R_{ab} b-coordinates rewritten as a-coordinates
the inner b cancels; read the product right to left

WORKED EXAMPLE

$R_{ab} = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$, $R_{bc} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, so $R_{ac} = \frac{1}{5} \begin{bmatrix} -4 & -3 \\ 3 & -4 \end{bmatrix}$ — still unit, still perpendicular.

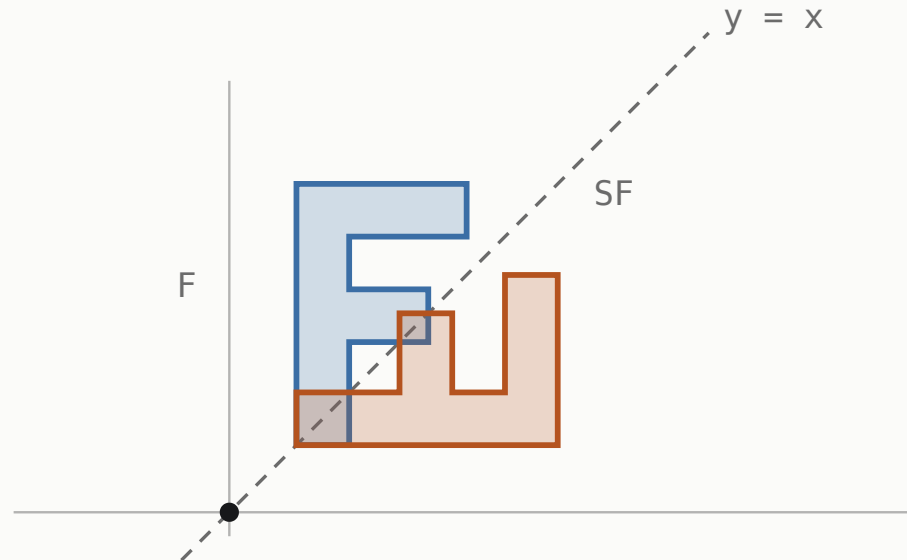


Rotations and Frames

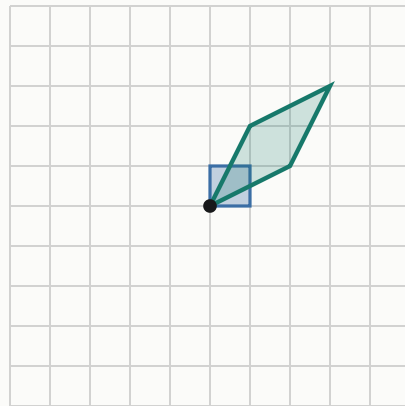
$$S = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad S^T S = I$$

S the mirror in the line $y = x$

$S^T S = I$ every length and every angle survives



The Determinant



THE AREA CHANGES

OR COLLAPSES

The Determinant

What number tells a turn from a mirror?

$$R \quad \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$$

$$S \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

EVERY LENGTH KEPT, EVERY ANGLE KEPT



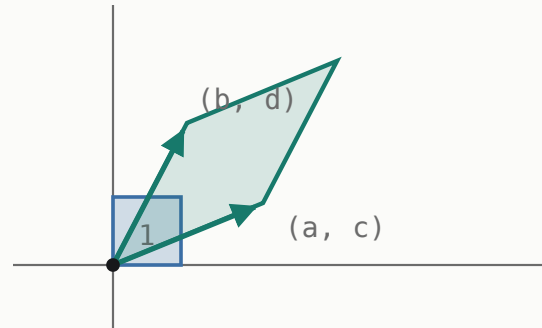
The Determinant

DEFINITION $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$

a, b, c, d the four entries, columns first

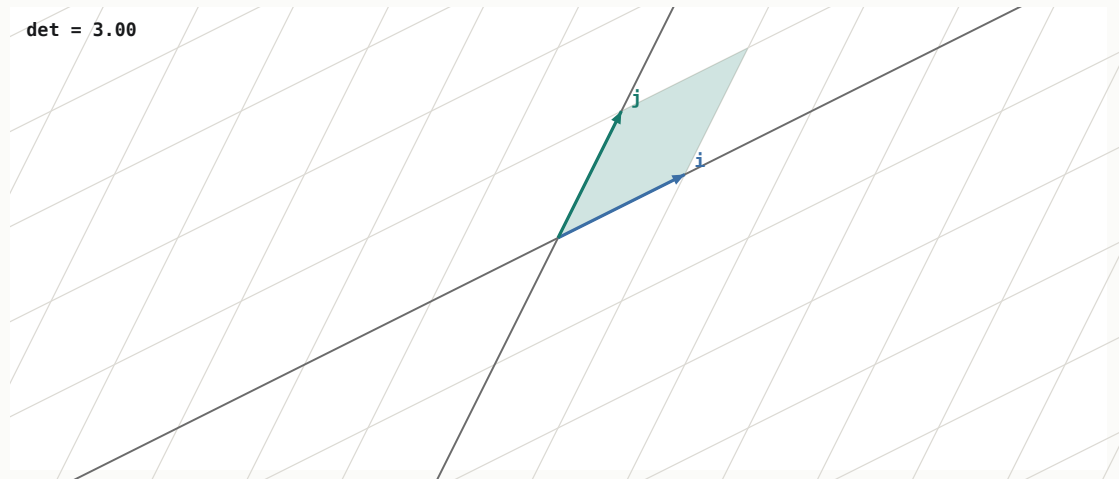
$ad - bc$ the signed area of the parallelogram the columns span

signed the sign records orientation, not size



Axler, S. *Linear Algebra Done Right*, 4th ed., 9.43 determinant of a matrix, p. 355; 9.47 the 2×2 formula, p. 356. Springer, 2024. Open access.

The Determinant



WORKED EXAMPLE

$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: $ad - bc = 2 \cdot 2 - 1 \cdot 1 = 3$, so every area in the plane is tripled.

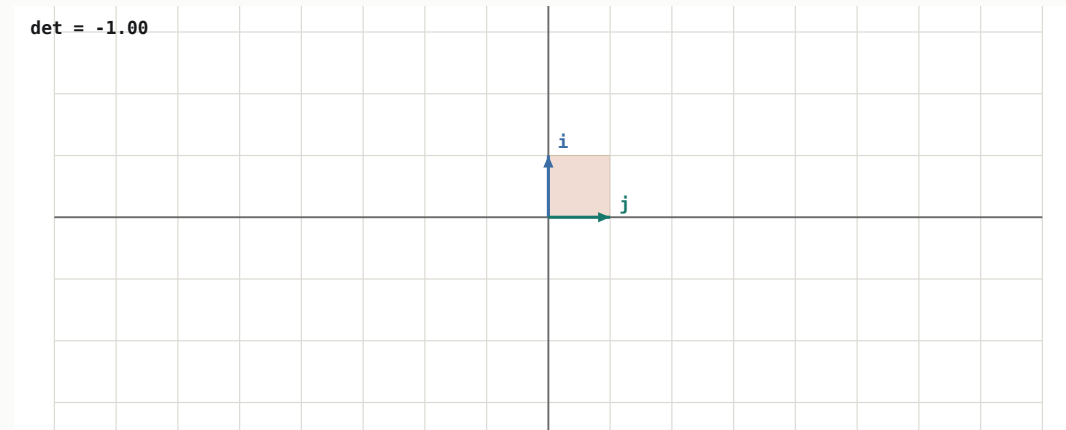
The Determinant

$$\det S = 0 \cdot 0 - 1 \cdot 1 = -1$$

$0 \cdot 0$ the diagonal product

$1 \cdot 1$ the off-diagonal product

-1 the plane has been turned over: a right-handed pair of axes is now left-handed



The square flattens, then reopens the other way round.

Sanderson, G. *Essence of Linear Algebra*, ch. 6: the determinant. 3Blue1Brown, 2016.

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

The Determinant

$$\det R = \frac{3}{5} \cdot \frac{3}{5} - \left(-\frac{4}{5}\right) \cdot \frac{4}{5} = \frac{9}{25} + \frac{16}{25} = 1$$

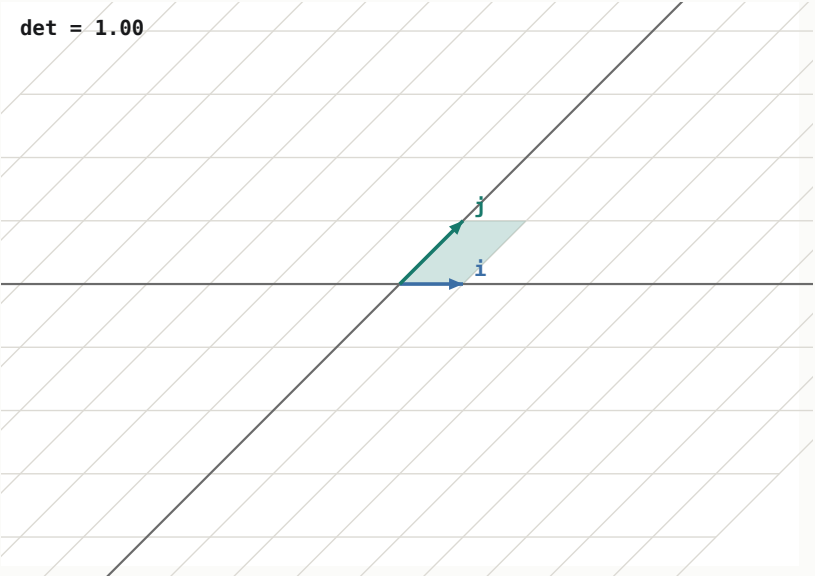
9/25 the diagonal product
 +16/25 minus a negative, so the off-diagonal term adds
 = 1 no stretch, no flip: a turn

MAP	$Q^T Q = I$	DET	VERDICT
rotation R	✓	+1	SO(2)
mirror S	✓	-1	reflection

Lynch, K. M. & Park, F. C. *Modern Robotics*, Definitions 3.1 and 3.2, p. 70. Cambridge, 2017.

The Determinant

RULE	WHY, IN VOLUME
$\det(AB) = \det A \cdot \det B$	do one map, then the other
$\det A^T = \det A$	same signed volume scale factor
$\det A^{-1} = 1/\det A$	undoing a scaling by 3 scales by 1/3
$\det(cA) = c^n \det A$	all n directions scaled by c
triangular T : $\det T = t_{11} \cdots t_{nn}$	shearing moves the box, not its volume



Row five, not row one: the other four rules are not drawn.

Axler, S. **Linear Algebra Done Right**, 4th ed., 9.48 upper-triangular, p. 356; 9.49 multiplicative, p. 357; 9.56(a) transpose, p. 360. Springer, 2024. Open access.

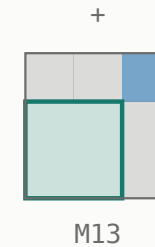
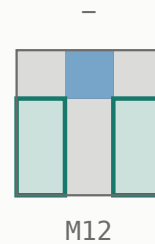
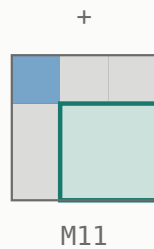
The Determinant

$$\det A = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$$

a_{1j} the three entries of the first row

M_{1j} the 2×2 determinant left when row 1 and column j are deleted

$+ - +$ a chequerboard of signs, starting at plus



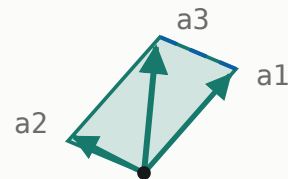
The Determinant

$$H = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

WORKED EXAMPLE

$$\det H = 1(1 \cdot 3 - 1 \cdot 1) - 0(0 \cdot 3 - 1 \cdot 2) + 1(0 \cdot 1 - 1 \cdot 2) = 2 - 0 - 2 = 0$$

Last week $a_3 = a_1 + a_2$ by eye; one number says the same thing.



NO THICKNESS

The Determinant

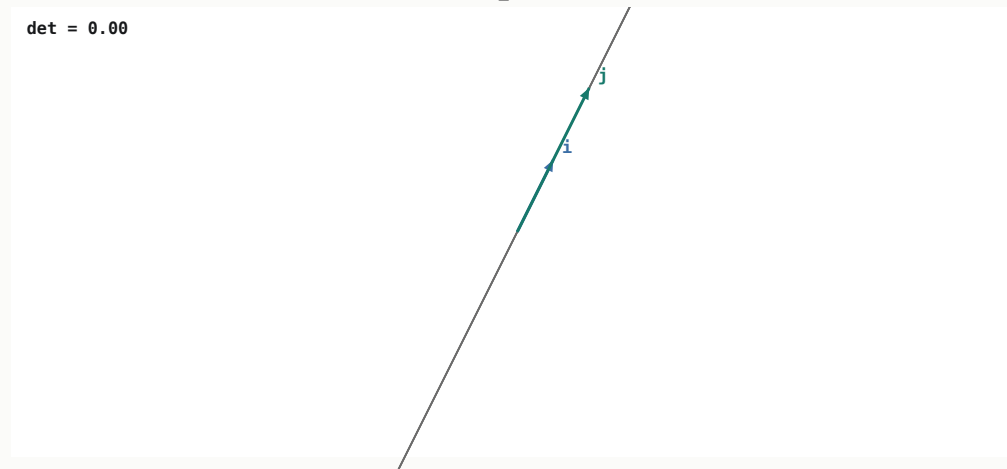
$$\det A = 0 \iff A \text{ is singular} \iff \mathcal{N}(A) \neq \{0\} \iff \text{rank } A < n$$

$\det A = 0$ the unit box is flattened

singular no inverse exists

$\mathcal{N}(A)$ the null space: some nonzero input is sent to zero

rank fewer than n independent columns



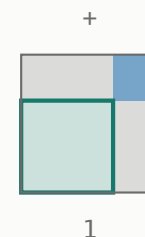
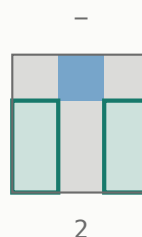
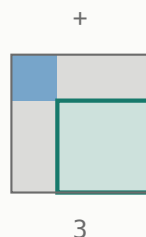
Every point of the plane lands on one line through the origin.

Axler, S. *Linear Algebra Done Right*, 4th ed., 9.50 invertible if and only if nonzero determinant, p. 357. Springer, 2024. Open access.

The Determinant

$$K = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \text{ singular?}$$

singular no inverse: some direction is destroyed



$\det K = 4$, so no. $2(4 - 1) - 1(2 - 0) + 0 = 6 - 2 = 4$, so $Kx = b$ has exactly one solution for every b . Or eliminate — subtracting a multiple of one row leaves det alone: $r_2 - \frac{1}{2}r_1$, then $r_3 - \frac{2}{3}r_2$, leaves pivots $2, \frac{3}{2}, \frac{4}{3}$, product 4.

The Determinant

$$\text{vol } A(\Omega) = |\det A| \cdot \text{vol } \Omega$$

Ω any region you like

$A(\Omega)$ its image under A

$|\det A|$ one factor, the same for every region

absolute value volume has no sign; orientation does



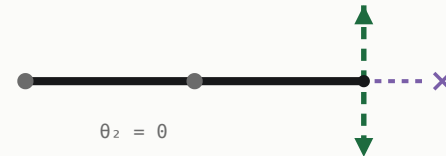
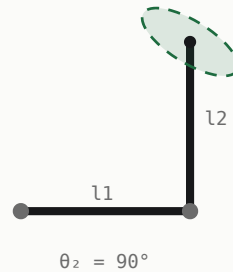
Axler, S. *Linear Algebra Done Right*, 4th ed., 9.61 T changes volume by a factor of $|\det T|$, p. 363. Springer, 2024. Open access.

The Determinant

$$\det J(\theta) = l_1 l_2 \sin \theta_2$$

- J joint rates to tip velocity, Lecture 1's Jacobian
- l_1, l_2 the two link lengths, both fixed
- θ_2 the elbow angle
- $= 0$ exactly when the elbow is straight or folded

tip velocities, $\| \text{joint rates} \| \leq 1$



WORKED EXAMPLE

$l_1 = l_2 = 1$: $\det J = \sin \theta_2$, and that set has area $\pi |\sin \theta_2| - \pi$ at 90° , and 0 when the elbow is straight.

Lynch, K. M. & Park, F. C. *Modern Robotics*, §5.3 singularities. Cambridge University Press, 2017.

The Determinant

$$\det(0.01 I_{10}) = (0.01)^{10} = 10^{-20}$$

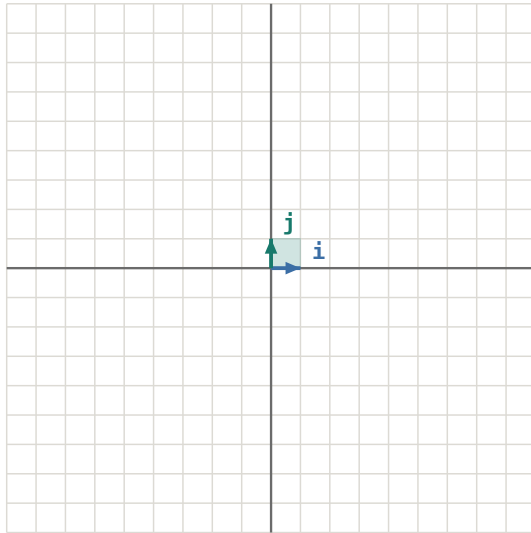
$0.01 I_{10}$ one hundredth of the identity in ten dimensions

$(0.01)^{10}$ one factor from each diagonal entry

10^{-20} small, and every direction still scaled alike

κ $\frac{\text{largest stretch}}{\text{smallest stretch}}$, the number to ask for — Lecture 4

det = 0.04



$0.2 I: \kappa = 1$

det = 0.04



$\text{diag}(1, 0.04): \kappa = 25$

The Determinant

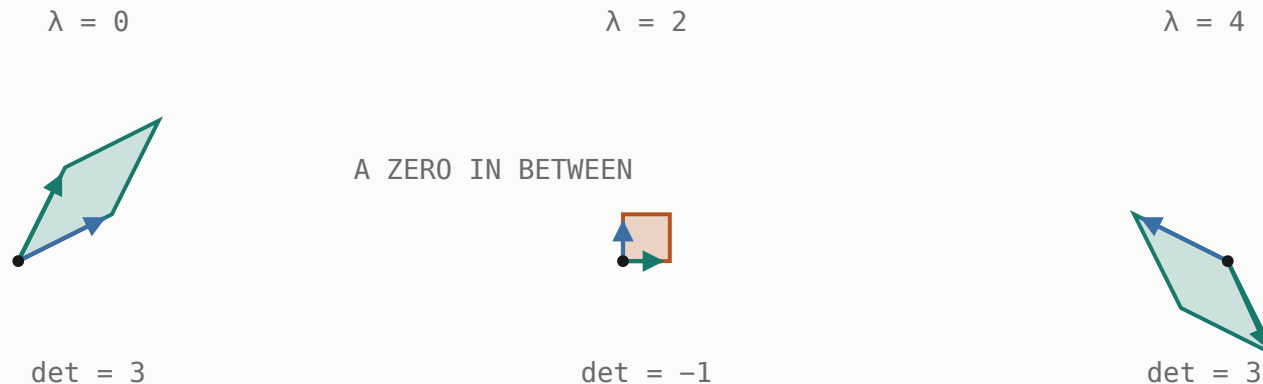
$$\det(A - \lambda I) = 0$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} : \quad \det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = (2 - \lambda)^2 - 1$$

$A - \lambda I$ A with λ subtracted along the diagonal

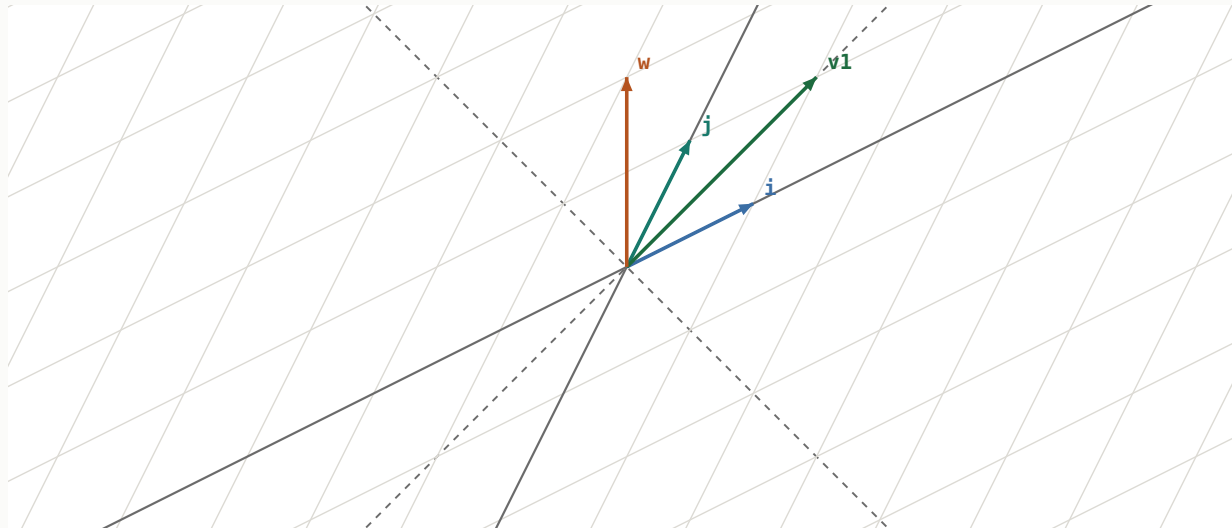
$\det = 0$ this chapter's test for collapse

λ unknown; the whole of the next chapter



Axler, S. *Linear Algebra Done Right*, 4th ed., 9.51, p. 358; 9.63 characteristic polynomial, p. 363. Springer, 2024. Open access.

Eigenvalues and Eigenvectors



One rider keeps its dashed line. The other does not.

Eigenvalues and Eigenvectors

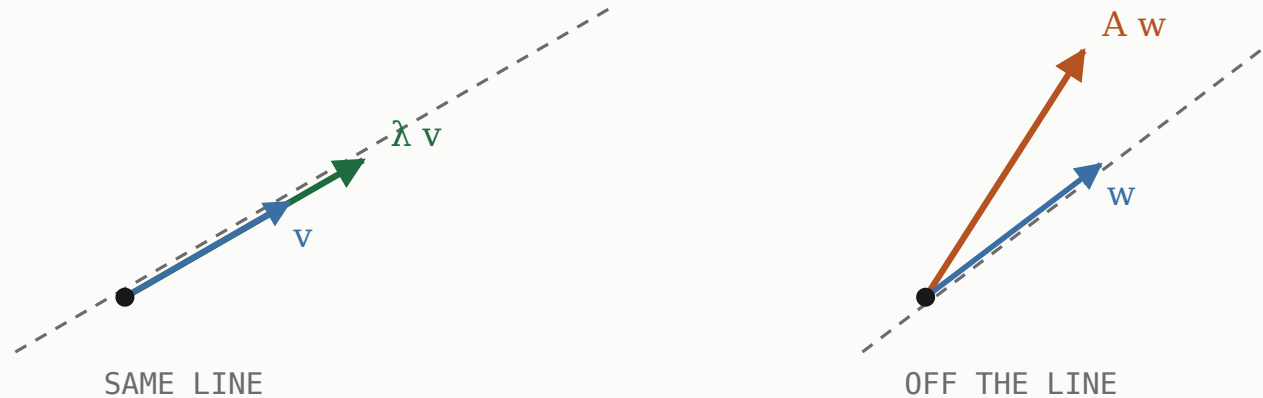
EIGENVECTOR $Av = \lambda v, \quad v \neq 0$

A the map: a square matrix

v the eigenvector, a direction A does not turn

λ the eigenvalue, the factor v is stretched by

$v \neq 0$ required; $\lambda = 0$ is allowed



Axler, S. *Linear Algebra Done Right*, 4th ed., 5.5 eigenvalue, p. 134; 5.8 eigenvector, p. 135. Springer, 2024. Open access.

Eigenvalues and Eigenvectors

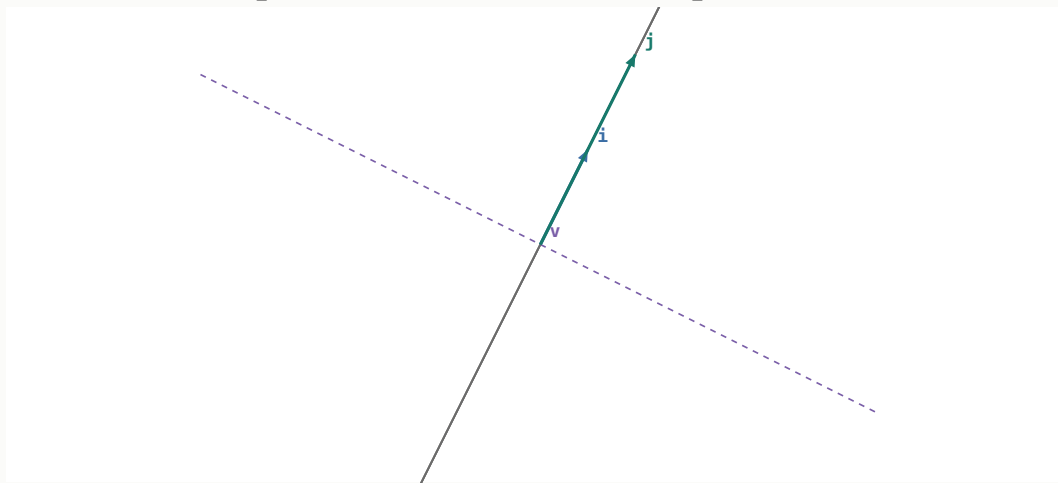
Allow $v = 0$ and every number is an eigenvalue of every matrix.

$$\lambda = 0 \text{ is an eigenvalue} \iff \mathcal{N}(A) \neq \{0\} \iff \det A = 0$$

$\lambda = 0$ $Av = 0$ for some $v \neq 0$

$\mathcal{N}(A)$ Lecture 1's null space: what the map destroys

$\det A = 0$ chapter 3's test for a collapse



The marked arrow goes to the origin.

Axler, S. *Linear Algebra Done Right*, 4th ed., 9.50 invertible if and only if nonzero determinant, p. 357. Springer, 2024. Open access.

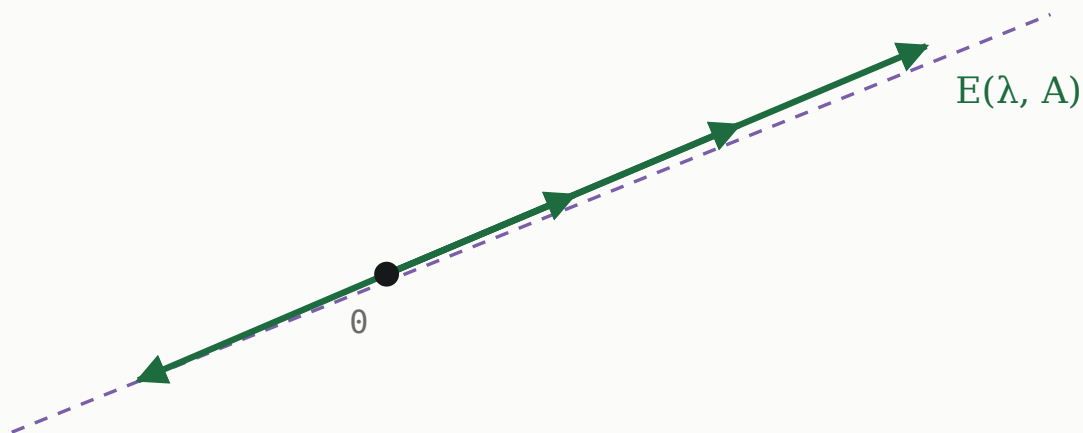
Eigenvalues and Eigenvectors

EIGENSPACE $E(\lambda, A) = \mathcal{N}(A - \lambda I) = \{x : Ax = \lambda x\}$

$E(\lambda, A)$ every vector A stretches by λ , and 0

$\mathcal{N}(A - \lambda I)$ Lecture 1's null space, of a different matrix

closed $A(\alpha x + \beta y) = \lambda(\alpha x + \beta y)$, so it is a subspace



Axler, S. *Linear Algebra Done Right*, 4th ed., 5.52 eigenspace, p. 164. Springer, 2024. Open access.

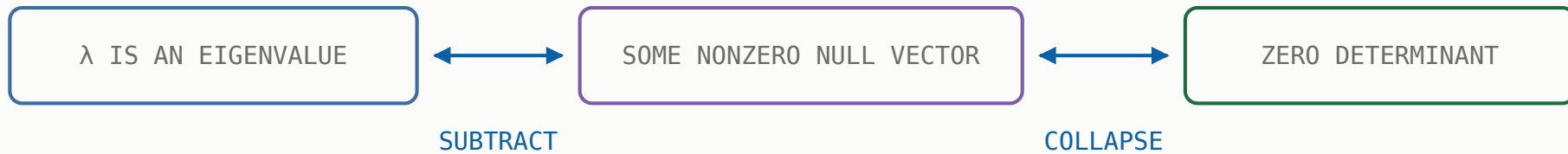
Eigenvalues and Eigenvectors

$$\lambda \text{ is an eigenvalue} \iff \exists v \neq 0 : (A - \lambda I)v = 0 \iff \det(A - \lambda I) = 0$$

subtract $\lambda v = \lambda Iv$, so both terms move to one side

$\exists v \neq 0$ some nonzero v solves it, not the v you name

collapse chapter 3: it has one exactly when its determinant is 0



Axler, S. *Linear Algebra Done Right*, 4th ed., 5.7 equivalent conditions, p. 135; 9.51 eigenvalue test, p. 358. Springer, 2024. Open access.

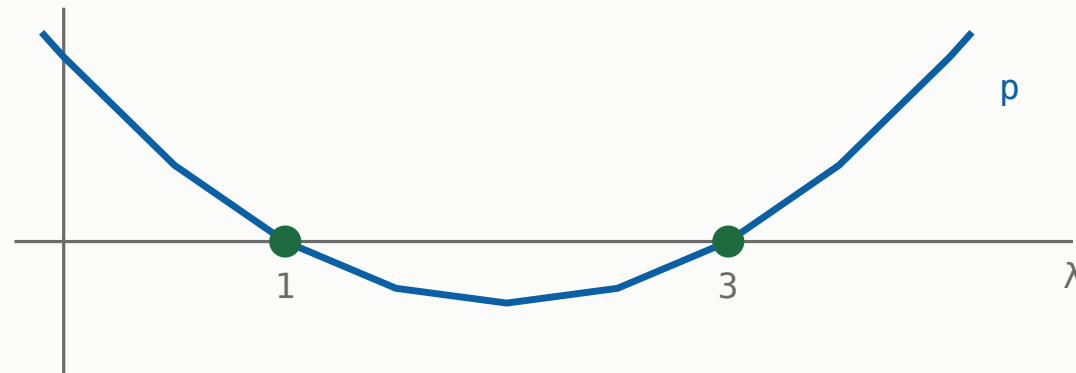
$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Eigenvalues and Eigenvectors

CHARACTERISTIC POLYNOMIAL $p(\lambda) = \det(A - \lambda I),$ $p = \underbrace{\lambda^2 - (\operatorname{tr} A)\lambda + \det A}_{2 \times 2 \text{ only}}$

p degree n in λ ; its roots are the eigenvalues

$\operatorname{tr} A$ the sum of the diagonal entries



Axler, S. *Linear Algebra Done Right*, 4th ed., 9.63 characteristic polynomial, p. 363; 8.47 trace of a matrix, p. 326. Springer, 2024. Open access.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Eigenvalues and Eigenvectors

WORKED EXAMPLE

$$A - \lambda I = \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$$

expand $(2 - \lambda)^2 = 4 - 4\lambda + \lambda^2$, then subtract 1

roots $\lambda = 3$ and $\lambda = 1$

check $\text{tr } A = 4$ and $\det A = 3$ give the same polynomial

$$\begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} \quad \begin{array}{l} \text{MAIN} \\ \text{minus} \\ \text{ANTI} \end{array}$$

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Eigenvalues and Eigenvectors

WORKED EXAMPLE

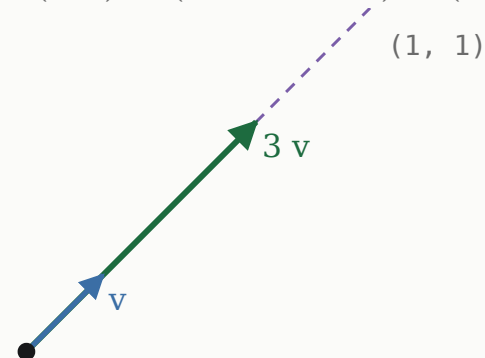
$\lambda = 3$: solve $(A - 3I)v = 0$, last week's null-space computation.

$$A - 3I = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$$

reduce $r_2 \rightarrow r_2 + r_1$, then $-r_1$: one pivot, one free

read $v_1 - v_2 = 0$, so $v_2 = t$ and $v = t(1, 1)$

check $A(1, 1) = (2 + 1, 1 + 2) = (3, 3) = 3(1, 1)$



Strang, G. **18.06 Linear Algebra**, Lecture 21: eigenvalues and eigenvectors. MIT OCW, 2010. CC BY-NC-SA 4.0.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Eigenvalues and Eigenvectors

WORKED EXAMPLE

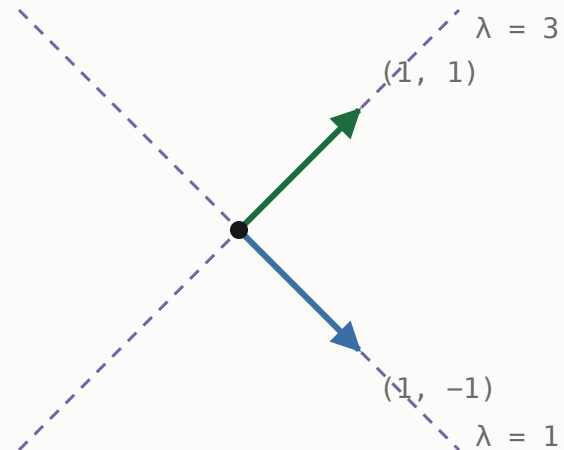
$\lambda = 1$: solve $(A - I)v = 0$.

$$A - I = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

reduce $r_2 \rightarrow r_2 - r_1$: one pivot, one free

read $v_1 + v_2 = 0$, so $v = t(1, -1)$

check $A(1, -1) = (2 - 1, 1 - 2) = (1, -1)$, unchanged



$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Eigenvalues and Eigenvectors

$$B^{-1}AB = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \quad Ab_1 = 3b_1, \quad Ab_2 = 1b_2$$

$b_1 = (1, 1)$ column 1 of B, and the first diagonal entry is 3

$b_2 = (1, -1)$ column 2 of B, and the second diagonal entry is 1

order swap the columns of B and the diagonal swaps with them

CHAPTER 1

THE BASIS THAT DIAGONALISED A

=

THIS CHAPTER

THE DIRECTIONS A DOES NOT TURN

Eigenvalues and Eigenvectors

$$\operatorname{tr} A = \lambda_1 + \cdots + \lambda_n, \quad \det A = \lambda_1 \cdots \lambda_n$$

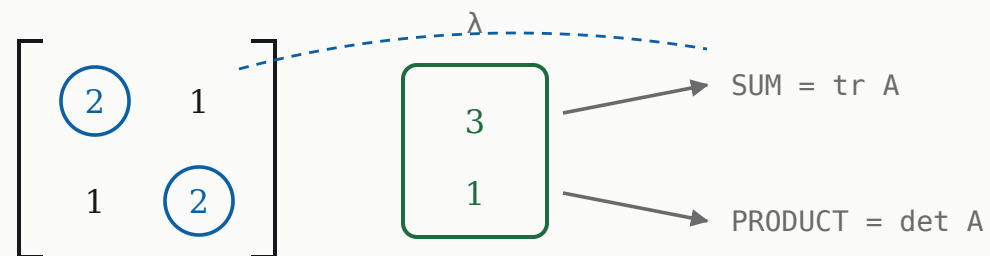
$\operatorname{tr} A$ the sum of the diagonal entries of A

λ_j the n roots of p , over $\mathbb{C}$ and with multiplicity

over $\mathbb{C}$ required: the next page is real and has no real root

WORKED EXAMPLE

$\operatorname{tr} A = 2 + 2 = 4 = 3 + 1$ and $\det A = 4 - 1 = 3 = 3 \cdot 1$.



Axler, S. **Linear Algebra Done Right**, 4th ed., 8.52 trace is the sum of the eigenvalues, p. 328; 9.55 the determinant is their product, p. 359. Springer, 2024. Open access.

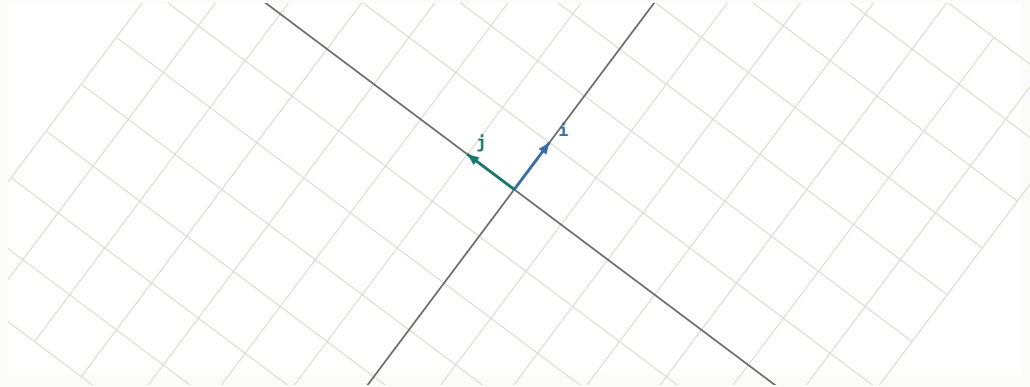
Eigenvalues and Eigenvectors

$$R = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \quad p(\lambda) = \lambda^2 - \frac{6}{5}\lambda + 1, \quad \Delta = \frac{36}{25} - 4 < 0$$

$$\text{tr } R = 6/5 \quad 3/5 + 3/5$$

$\det R = 1$ a rotation preserves area

$\Delta = b^2 - 4ac$ of $a\lambda^2 + b\lambda + c$; negative here, so no real root and no real direction survives



No dashed line appears.

Lynch, K. M. & Park, F. C. **Modern Robotics**, Definition 3.2, SO(2), p. 70. Cambridge University Press, 2017.

Eigenvalues and Eigenvectors

What are the eigenvalues of $\begin{pmatrix} 3 & 7 \\ 0 & 5 \end{pmatrix}$?

3 and 5: $\det(A - \lambda I) = (3 - \lambda)(5 - \lambda)$, and a triangular determinant is the product of the diagonal.

For $\lambda = 3$, $v = (1, 0)$. For $\lambda = 5$ it is $(7, 2)$, not $(0, 1)$: $(A - 5I)(7, 2) = (-14 + 14, 0) = 0$.

$$\begin{bmatrix} \textcircled{3} & 7 \\ 0 & \textcircled{5} \end{bmatrix} \quad \begin{array}{l} \lambda = 3, 5 \\ v = (1, 0) \text{ and } (7, 2) \end{array}$$

Axler, S. *Linear Algebra Done Right*, 4th ed., 9.48 determinant of an upper-triangular matrix, p. 356. Springer, 2024. Open access.

Eigenvalues and Eigenvectors

SIMILAR MATRICES $Av = \lambda v \implies (P^{-1}AP)(P^{-1}v) = \lambda (P^{-1}v)$

P columns: the new basis in old coordinates, so $[x]_{\text{old}} = P[x]_{\text{new}}$

$P^{-1}v$ the same eigenvector, read in the new coordinates

λ untouched: similar matrices share their eigenvalues

$$\begin{array}{c} A \\ \left[\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right] \\ \{3, 1\} \end{array}$$

$$\begin{array}{c} C^{-1}AC \\ \left[\begin{array}{cc} 1 & 0 \\ 1 & 3 \end{array} \right] \\ \{3, 1\} \end{array}$$

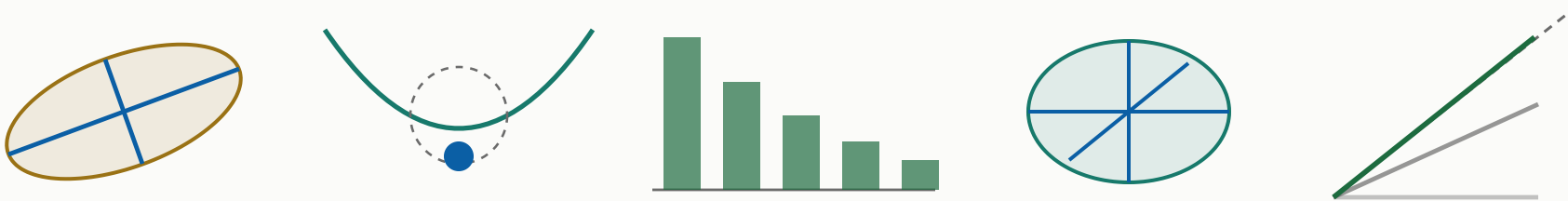
$$\begin{array}{c} B^{-1}AB \\ \left[\begin{array}{cc} 3 & 0 \\ 0 & 1 \end{array} \right] \\ \{3, 1\} \end{array}$$

Strang, G. **18.06 Linear Algebra**, Lecture 28: similar matrices and Jordan form. MIT OCW, 2010. CC BY-NC-SA 4.0.

Eigenvalues and Eigenvectors

- Covariance: axes and spread of an ellipse — Lecture 8, PCA in Lecture 4
- Loss curvature $\frac{1}{2}x^T H x$: the largest eigenvalue caps the step — Lecture 6
- A linear iteration is stable or it is not — next chapter, Lecture 7
- Principal moments of an inertia matrix
- Power iteration: the turning arrows as an algorithm, one $|\lambda|$ strictly largest

$cf, c > 0$ same minimiser, gradient $\times c$



$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

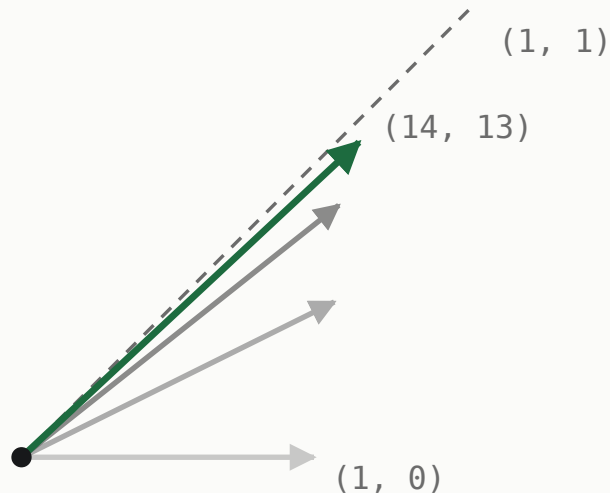
Eigenvalues and Eigenvectors

WORKED EXAMPLE

$x_0 = (1, 0)$: $Ax_0 = (2, 1)$, $A^2x_0 = (5, 4)$, $A^3x_0 = (14, 13)$.

Divide by the first entry: $(1, 0)$, $(1, \frac{1}{2})$, $(1, \frac{4}{5})$, $(1, \frac{13}{14})$.

First entries grow by $2, \frac{5}{2}, \frac{14}{5}$ toward 3, the strictly largest $|\lambda|$. From $(1, -1)$ nothing turns.



Directions only; each arrow is drawn to its own scale.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Eigenvalues and Eigenvectors

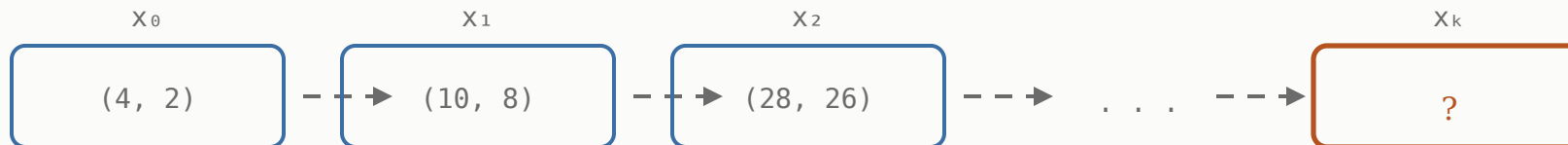
Two directions and two factors. Not yet: the hundredth state without a hundred products.

$$x_{k+1} = Ax_k \implies x_k = A^k x_0$$

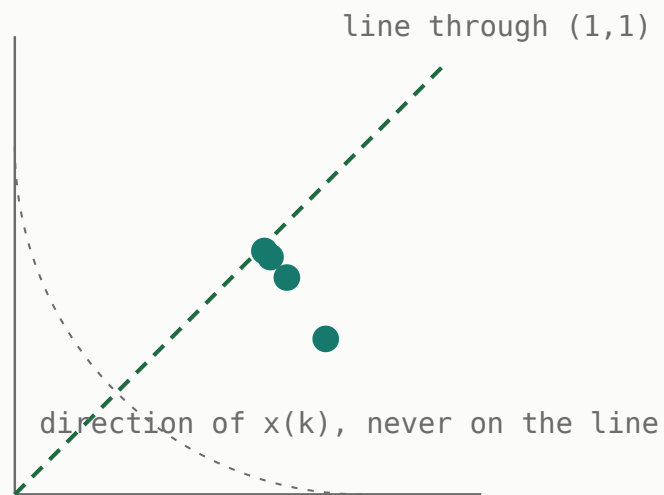
x_k the state after k steps

build A^k one at a time, $k - 1$ products, about n^3 each

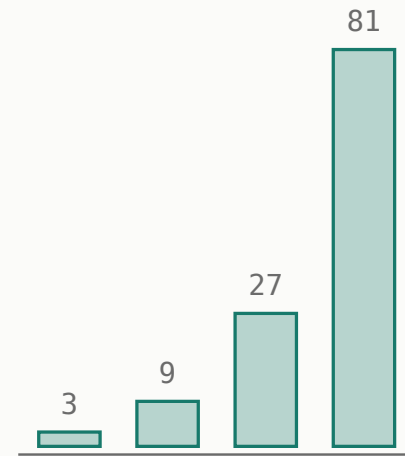
only $A^k x_0$ k matrix-vector products, about n^2 each



Diagonalization and Modes



$\lambda = 3$



$\lambda = 1$

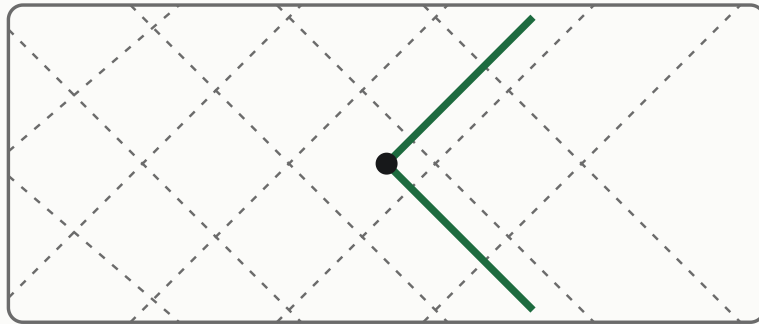


Diagonalization and Modes

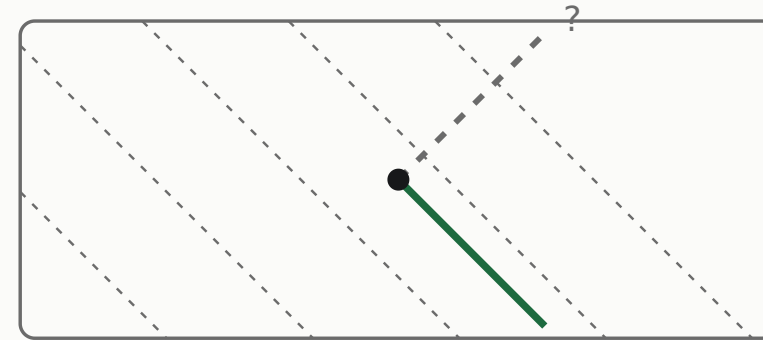
DIAGONALIZABLE A is diagonalizable over $\mathbb{R} \iff \mathbb{R}^n$ has a basis of eigenvectors of A

a basis of eigenvectors n independent directions, each one only stretched by A

the consequence in that basis the matrix of the operator is diagonal



TWO DIRECTIONS: A BASIS



ONE DIRECTION: NOT A BASIS

Axler, S. **Linear Algebra Done Right**, 4th ed., 5.50 diagonalizable, p. 163; 5.55 conditions equivalent to diagonalizability, p. 165. Springer, 2024. Open access.

Diagonalization and Modes

$$AV = V\Lambda \implies A = V\Lambda V^{-1}$$

- V the eigenvectors as columns, invertible because they are a basis
- Λ the eigenvalues on the diagonal, in the same order as those columns

step two multiply both sides on the right by V^{-1}

$$\begin{array}{c} A \quad V \\ \left[\begin{array}{|c|c|} \hline A \ v1 & A \ v2 \\ \hline \end{array} \right] = \left[\begin{array}{|c|c|} \hline \lambda1 \ v1 & \lambda2 \ v2 \\ \hline \end{array} \right] \\ \text{-----} \quad \text{column } j \text{ of both is } \lambda_j \ v_j \quad \text{-----} \end{array}$$

Goodfellow, I., Bengio, Y. & Courville, A. **Deep Learning**, ch. 2.7 eigendecomposition. MIT Press, 2016.

Diagonalization and Modes

WORKED EXAMPLE

$$V\Lambda = \begin{bmatrix} 3 & 1 \\ 3 & -1 \end{bmatrix}, \quad (V\Lambda)V^{-1} = \frac{1}{2} \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} = A$$

$V\Lambda$ column j of V multiplied by λ_j , which is 3 and 1
 the half $V^{-1} = \frac{1}{2}V$ for this V , so it rides to the end

$$A = \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}_V \underbrace{\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}}_{\Lambda, \text{ DIAGONAL}} \underbrace{\frac{1}{2}}_{V^{-1}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

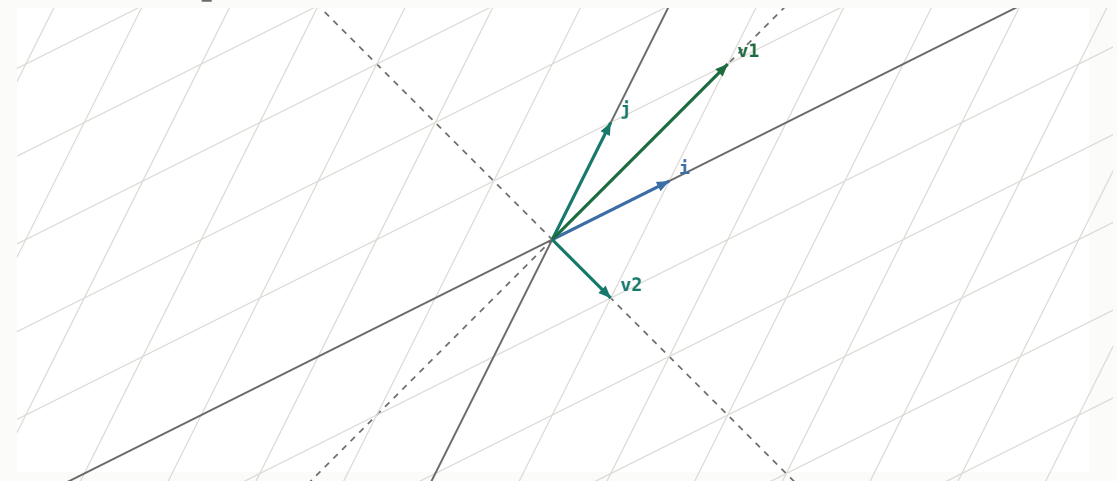
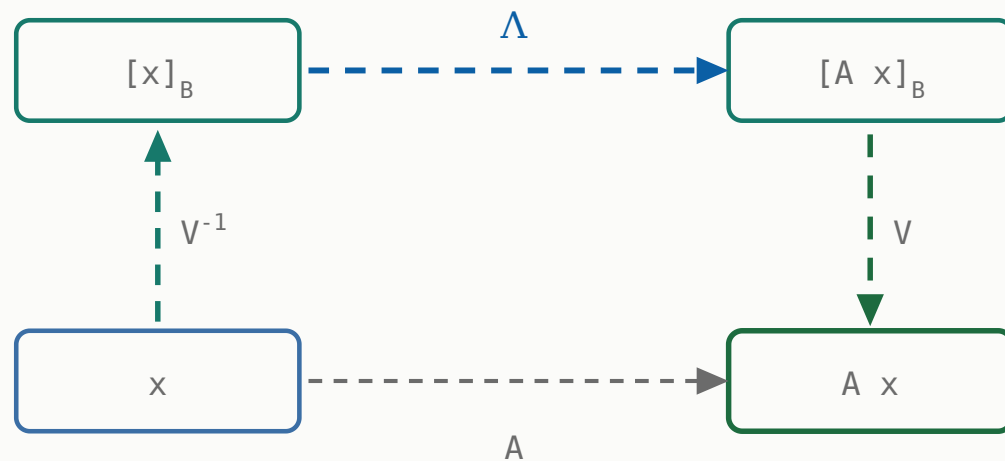
Diagonalization and Modes

CHANGE OF BASIS $\Lambda = V^{-1}AV = [A]_{\mathcal{B}}$

$V^{-1}AV$ similarity, run with the eigenvectors as the new basis

$[A]_{\mathcal{B}}$ the matrix of the operator in the basis $\mathcal{B}$

Λ one operator, three names; this V is p60's basis matrix B



The dashed lines are the columns of V — the axes Λ is written in.

Diagonalization and Modes

$$A^k = V \Lambda^k V^{-1}, \quad \Lambda^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k)$$

the middle $V^{-1}V = I$ cancels $k - 1$ times

Λ^k raise each diagonal entry; nothing else moves

the cost after the factorization, all dependence on k is n scalar powers

$$A \cdot A \cdot A = V \quad \Lambda \quad \begin{array}{c} \diagup \quad \diagdown \\ V^{-1} \quad V \\ \diagdown \quad \diagup \end{array} \quad \Lambda \quad \begin{array}{c} \diagup \quad \diagdown \\ V^{-1} \quad V \\ \diagdown \quad \diagup \end{array} \quad \Lambda \quad V^{-1}$$

$= I \qquad \qquad \qquad = I$

$$= V \quad \Lambda^3 \quad V^{-1} \quad \text{two factors survive, whatever } k \text{ is}$$

Strang, G. **18.06 Linear Algebra**, Lecture 22, diagonalization and powers of A. MIT OpenCourseWare, 2010. CC BY-NC-SA 4.0.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

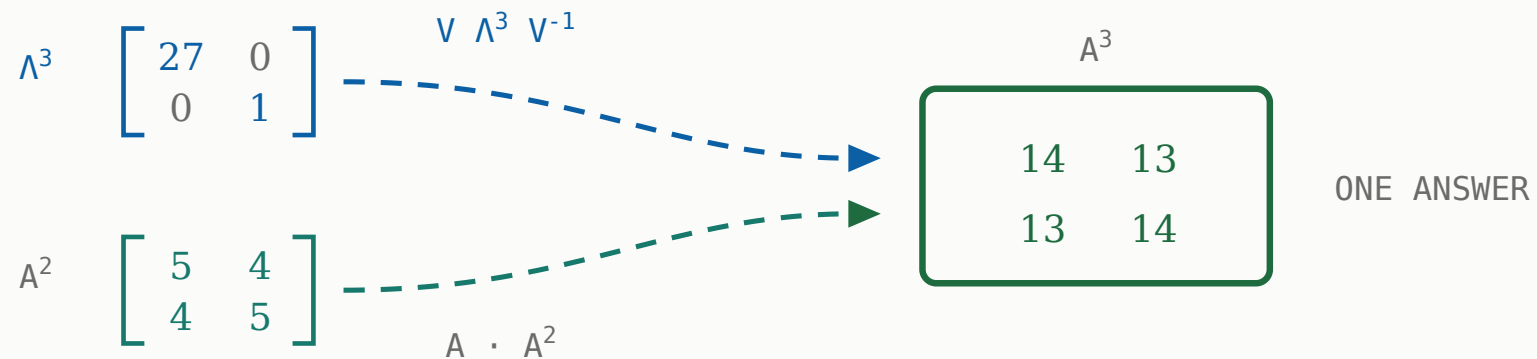
Diagonalization and Modes

WORKED EXAMPLE

$$A^k = V \Lambda^k V^{-1} = \frac{1}{2} \begin{bmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{bmatrix}$$

$k = 3$ $3^3 = 27$, so $(27 + 1)/2 = 14$ and $(27 - 1)/2 = 13$

check square A, then multiply by A once more



$$V = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Diagonalization and Modes

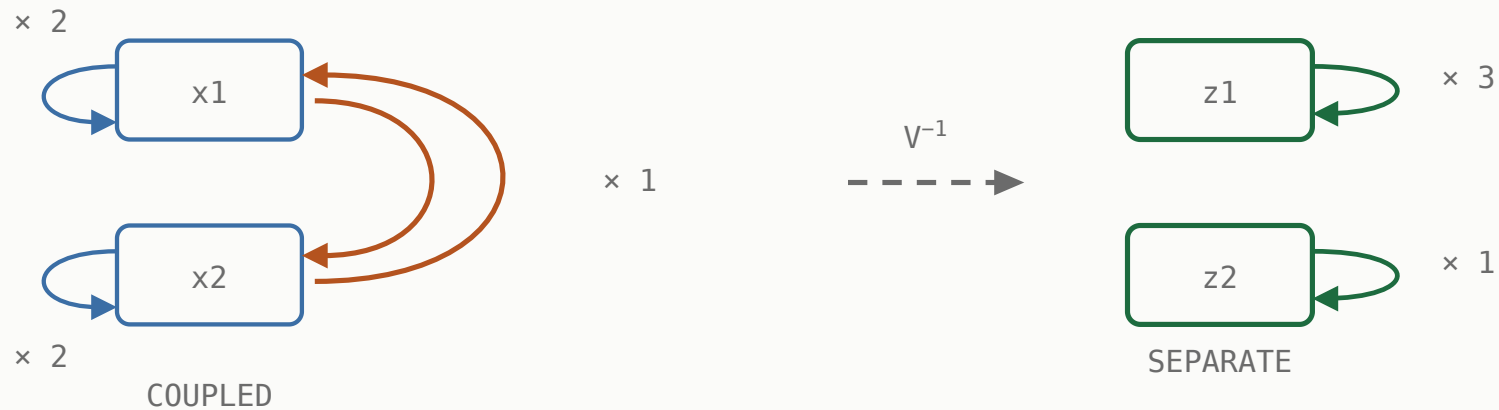
$$x_{k+1} = Ax_k, \quad z_k = V^{-1}x_k$$

MODAL COORDINATES

$$z_{k+1} = V^{-1}Ax_k = (V^{-1}AV)z_k = \Lambda z_k \implies z_k = \Lambda^k z_0$$

z the state in eigen-coordinates, one number per mode

$\Lambda = \text{diag}(3, 1)$ no coupling: row i mentions only z_i



Strang, G. **18.06 Linear Algebra**, Lecture 22, diagonalization and powers of A . MIT OpenCourseWare, 2010. CC BY-NC-SA 4.0.

$$\Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

Diagonalization and Modes

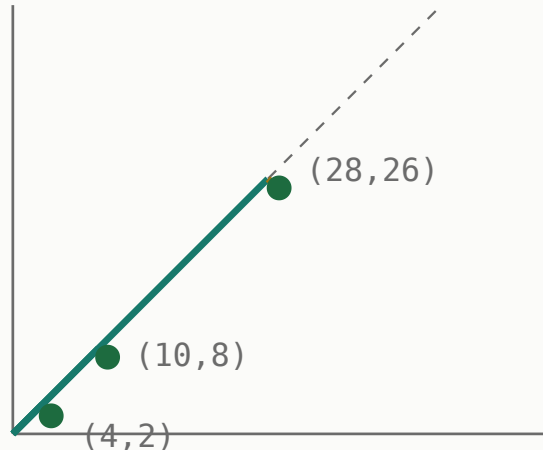
WORKED EXAMPLE

$$x_k = V\Lambda^k z_0 = 3 \cdot 3^k \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \cdot 1^k \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$z_0 = (3, 1)$ $V^{-1}x_0$, the coordinates already found for $x_0 = (4, 2)$

$k = 1, 2$ $9(1, 1) + (1, -1) = (10, 8)$, $27(1, 1) + (1, -1) = (28, 26)$

every k the entries differ by 2: the dots sit on one straight line



— $\lambda = 3$ part: 3, 9, 27

— $\lambda = 1$ part: never moves

Diagonalization and Modes

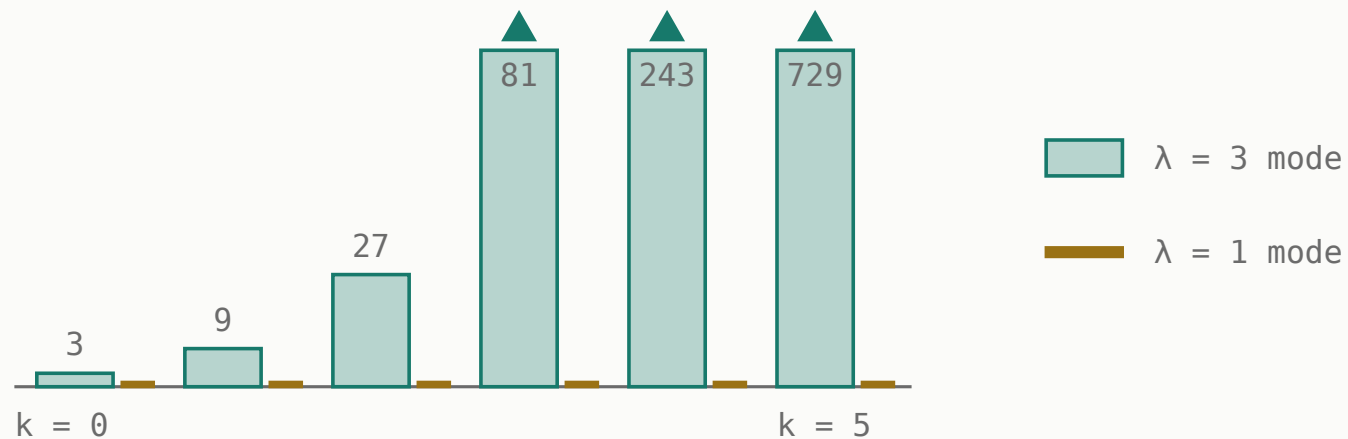
$$x_k = \sum_j c_j \lambda_j^k v_j, \quad \lambda_1^{-k} x_k \longrightarrow c_1 v_1$$

λ_1 strictly largest in absolute value, and nonzero

$c_1 \neq 0$ x_0 must carry some v_1

residual here $x_k - 3 \cdot 3^k(1,1) = (1,-1)$ at every k

at $k = 10$ $3 \cdot 3^{10} = 177,147$ against 1



Diagonalization and Modes

$$\rho(A) = \max_j |\lambda_j|; \quad \text{diagonalizable } A : A^k x_0 \rightarrow 0 \quad \forall x_0 \iff \rho(A) < 1$$

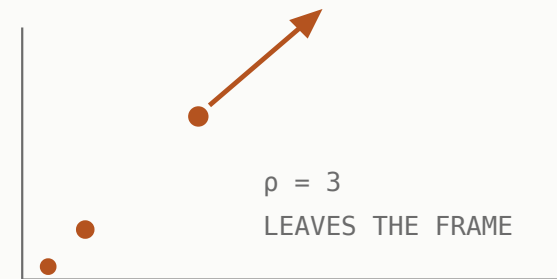
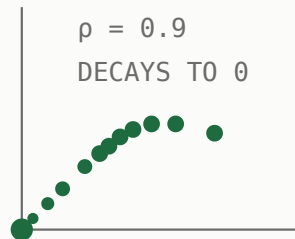
ρ the largest of the $|\lambda_j|$, over every eigenvalue

x_0 any starting state

strict $\rho = 1$ is not enough

WORKED EXAMPLE

A_d has 0.7 on the diagonal and 0.2 off it: the same eigenvectors, and eigenvalues 0.9 and 0.5.



Strang, G. **18.06 Linear Algebra**, Lecture 22, diagonalization and powers of A. MIT OpenCourseWare, 2010. CC BY-NC-SA 4.0.

$$\text{Sh} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

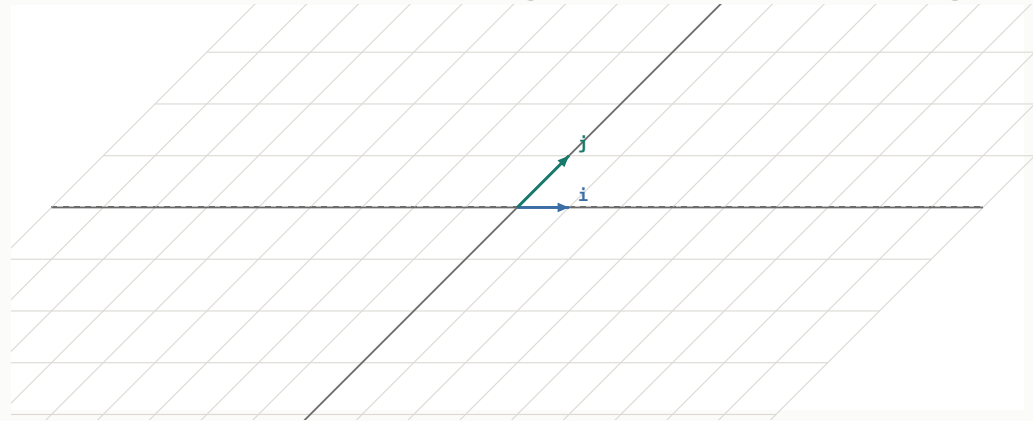
Diagonalization and Modes

DEFECTIVE $(\text{Sh} - I)v = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} v = 0 \Rightarrow v_2 = 0 \Rightarrow v \in \text{span}\{(1, 0)\}$

Sh the shear in the corner: x slides by y

$\lambda = 1$ twice from $(1 - \lambda)^2$, yet one direction solves it

defective short of a basis of eigenvectors, so not diagonalizable



One dashed invariant line, where two would have made a basis.

Axler, S. *Linear Algebra Done Right*, 4th ed., 5.55 conditions equivalent to diagonalizability, p. 165. Springer, 2024. Open access.

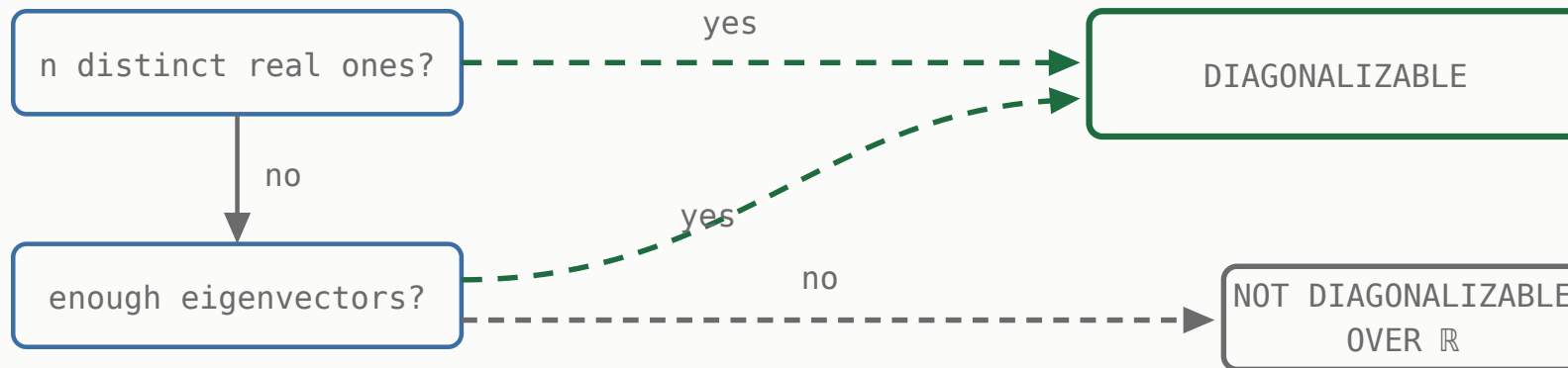
Diagonalization and Modes

n distinct real eigenvalues $\Rightarrow A$ is diagonalizable over $\mathbb{R}$

real slide 62's rotation: distinct complex roots, no real basis

one way only the identity repeats an eigenvalue, yet is diagonal

Over $\mathbb{C}$ every square matrix has a Jordan form: diagonal plus ones just above it.



Axler, S. **Linear Algebra Done Right**, 4th ed., 5.58, p. 166. Springer, 2024. Jordan form quoted without proof: Horn, R. & Johnson, C. **Matrix Analysis**, 2nd ed. Cambridge, 2012.

Diagonalization and Modes

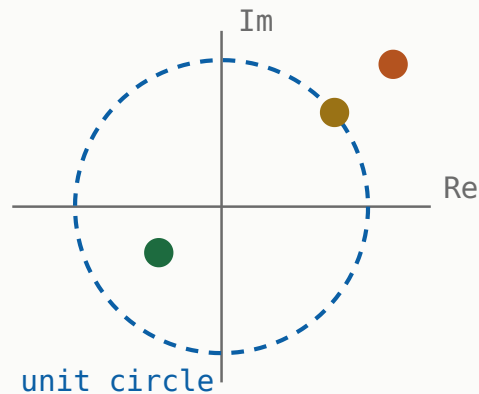
$$x_{k+1} = (A - BK)x_k \rightarrow 0 \quad \forall x_0 \iff \rho(A - BK) < 1$$

A, B the plant; this B is the input matrix, not the basis matrix of p10-24

K the gain you design; it moves the eigenvalues

$\rho < 1$ strictly inside; a repeated root on the circle can still grow

next door Lecture 7 does $\operatorname{Re} \lambda < 0$; Lecture 8 pushes a covariance through A



The five questions, answered

1 · COORDINATES

what are these numbers
written in?

2 · ROTATIONS

which change of basis
is a turn?

3 · DETERMINANT

does it collapse,
and does it flip?

4 · EIGENVECTORS

which directions
survive?

5 · DIAGONALIZATION

what is A to the k ?

Courses and books

1. Axler, S. [*Linear Algebra Done Right*](#), 4th ed. Springer, 2024. Open access. Quotes include 3.84, 5.5, 5.52, 9.50, 9.61.
2. Strang, G. [*18.06 Linear Algebra*](#), L19, 20, 21, 22, 28, 31. MIT OCW, 2010. CC BY-NC-SA 4.0.
3. Lynch and Park. [*Modern Robotics*](#). Cambridge, 2017.
4. Grizzle, J. [*ROB 501*](#), notes 11. Michigan, 2018. CC BY-NC 4.0.
5. Sanderson, G. [*Essence of Linear Algebra*](#), ch. 6. 3Blue1Brown, 2016.

Carried forward

6. Goodfellow, Bengio and Courville. [*Deep Learning*](#), ch. 2.7, 2.8. MIT Press, 2016.
7. Horn and Johnson. *Matrix Analysis*, 2nd ed. Cambridge, 2012. Jordan form, slide 80.
8. Strang, G. *Linear Algebra and Learning from Data*. Wellesley-Cambridge, 2019. Lectures 3, 4.
9. ROB-GY 6013. *Lecture 1, Introduction and Linear Algebra I*. Null space and rank.
10. [*Notation and Conventions*](#) · [*Textbooks and Reading Guide*](#).